



CBSE SKILL EDUCATION • SUBJECT CODE 417

Artificial Intelligence

Premium Printable Textbook & Study Guide

CLASS IX

Session 2025–2026 • As per Latest CBSE Curriculum

Complete Theory • Core Diagrams • Solved Flowcharts • Comparative Tables

Step-by-Step Python Guide • 24 Solved Programs • Viva Voce Preparation

One-Shot Summaries • Key terms • Solved MCQs, Short & Long Questions

A4 Printable Textbook Format (Ctrl + P → Save as PDF)

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Communication Skills-I

Communication is the lifeblood of human relationship and collaboration. This unit introduces the communication cycle, the elements that make it effective, perspectives and barriers that affect understanding, and the methods (verbal, non-verbal, and visual) we use to share meaning.

1.1 Introduction to Communication & The Communication Cycle

The word **communication** originates from the Latin word *communicare*, which means "to share." It is the process of exchanging information, ideas, thoughts, feelings, and emotions between two or more individuals.

FORMAL DEFINITION

Communication is a two-way process of sharing information to reach a common understanding. It is complete only when the receiver has understood the sender's message as intended and provided feedback.

The Communication Cycle

Communication is not a single, isolated act; it is a cyclic process consisting of several sequential components:

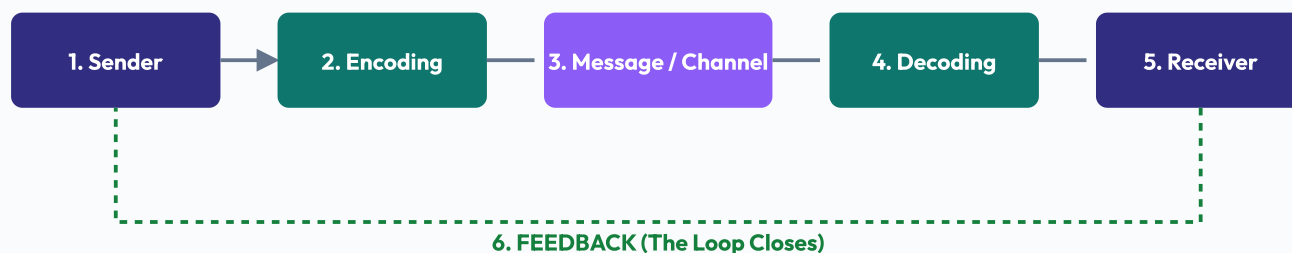


FIGURE 1.1: THE COMMUNICATION CYCLE ELEMENTS & FEEDBACK LOOP

1. **Sender:** The source who conceives the idea or information and initiates the communication.
2. **Message:** The actual content, idea, or information to be transmitted.
3. **Encoding:** The process of converting the idea into symbolic form (words, symbols, gestures) suitable for transmission.
4. **Channel / Medium:** The physical vehicle through which the encoded message travels (e.g., face-to-face talk, email, SMS, phone call).
5. **Receiver:** The target individual or group for whom the message is intended.
6. **Decoding:** The receiver's interpretation and translation of the encoded message back into understandable thoughts.
7. **Feedback:** The receiver's response or reaction sent back to the sender. This confirms whether the message was received and understood.

REAL-WORLD EXAMPLE

Teacher: "Please submit your AI practical files by Friday." (Sender encodes the Message and transmits via verbal Medium).

Student: Nods and says, "Yes, sir, we will submit them on Friday morning." (Receiver decodes message and sends verbal/non-verbal Feedback).

1.2 Barriers to Effective Communication

An effective communication cycle can be disrupted by **barriers**—obstacles that result in misunderstanding, distortion, or complete failure of communication.

BARRIER TYPE	DESCRIPTION	MITIGATION STRATEGY
Physical / Environmental	Distances, background noise, poor network, faulty equipment, or physical walls.	Choose a quiet venue; ensure hardware/network is functional before starting.
Language / Linguistic	Differences in languages, dialects, complex jargon, or accent differences.	Use simple language, avoid jargon, and seek translators or clarification.
Cultural	Varying interpretations of gestures, values, body language, and space between different cultures.	Educate oneself on cultural norms; avoid making generalizations or assumptions.
Interpersonal / Psychological	Personal biases, emotional states (anger, stress), lack of interest, or mistrust of the sender.	Communicate with an open mind; manage emotions before responding; practice active listening.
Organizational	Rigid hierarchy, complex procedures, rules, and lack of transparency within a school or office.	Establish clear communication channels and simplified reporting lines.

1.3 Perspectives in Communication

Our **perspectives** act as personal filters that color how we view the world and interpret messages. These perspectives are formed by various factors:

- **Visual Perception:** The brain's tendency to interpret pictures or situations based on quick visual cues or prior familiarity.
- **Past Experiences:** Letting previous events dictate how we interact. If a teacher was angry last Monday, we might assume they are angry today when they ask a question.
- **Prejudices & Stereotypes:** Preconceived notions about individuals based on gender, race, age, or background, which lead to biased listening.
- **Feelings & Emotions:** Current state of mind. An anxious person may perceive a constructive suggestion as a harsh attack.
- **Environment:** The surrounding setting. A conversation at home feels different from the same conversation in a strict principal's office.

EXAM TIP: THE 7 CS OF EFFECTIVE COMMUNICATION

To overcome barriers and perspective filters, construct your message following the **7 Cs**:

1. **Clear:** Use plain language; focus on one main point per sentence.
2. **Concise:** Avoid unnecessary words; keep it brief and straight to the point.
3. **Concrete:** Use facts and figures instead of vague generalizations.
4. **Correct:** Ensure spelling, grammar, and facts are accurate.
5. **Coherent:** Organize ideas logically; ensure they flow naturally.
6. **Complete:** Provide all necessary information that the receiver needs to take action.
7. **Courteous:** Respect the receiver's time and feelings; use a polite tone.

1.4 Methods of Communication

We communicate using three primary channels: Verbal, Non-verbal, and Visual.

1. Verbal Communication

Verbal communication uses spoken or written words to share information.

- **Oral/Spoken Communication:** Face-to-face discussions, phone calls, online meetings. Advantages: Quick feedback, emotional connection, permits voice modulation. Disadvantages: No permanent record, prone to misinterpretation in noise.

- **Written Communication:** Emails, letters, books, WhatsApp messages. Advantages: Permanent record, allows editing before sending, suitable for complex data. Disadvantages: Delays feedback, requires literacy and time to write.

2. Non-Verbal Communication

Non-verbal communication is the exchange of messages without words, relying instead on physical actions and cues:

- **Kinesics (Body Movements):** Head nods, posture (slouched vs. upright), hand gestures.
- **Facial Expressions:** Smiles, frowns, raised eyebrows, which directly convey emotions.
- **Eye Contact (Oculesics):** Shows attentiveness, sincerity, and interest when maintained.
- **Paralanguage:** Tone of voice, speed, pitch, and pauses during speech.
- **Proxemics (Physical Space):** Intimate, personal, social, or public distance kept between people.

3. Visual Communication

Visual communication involves transmitting ideas through visual aids like diagrams, signs, posters, logos, and charts. It transcends language barriers and is instantly recognizable.

COMMON VISUAL SYMBOLS

- **Red Circle with a Slash:** Indicates prohibition (e.g., No Parking, No Smoking).
- **Triangular Yellow Sign:** Represents warnings (e.g., Danger, Slippery Road).
- **Green Running Figure:** Represents emergency exit.

UNIT 1 KEY VOCABULARY

- **Communicare:** Latin root meaning to share or make common.
- **Feedback:** The response that tells the sender if the message was received accurately.
- **Jargon:** Technical terms specific to a profession, which can act as a barrier to outsiders.
- **Paralanguage:** Non-verbal cues of the voice (pitch, volume, rate) that accompany spoken language.

Unit 1 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. Which of the following is the correct order of the communication cycle?

- a) Sender → Decoding → Message → Channel → Feedback
- b) Sender → Encoding → Message → Channel → Decoding → Receiver → Feedback
- c) Receiver → Message → Decoding → Sender → Feedback
- d) Sender → Message → Receiver → Feedback → Channel

Answer: b

Q2. An warning sign on a road (like a school ahead sign) is an example of which type of communication?

- a) Verbal Oral
- b) Non-verbal paralanguage
- c) Visual
- d) Written

Answer: c

Q3. Which "C" of communication states that we should use exact facts and figures rather than vague ideas?

- a) Concise
- b) Concrete
- c) Correct
- d) Coherent

Answer: b

Short Answer Questions (2-3 Marks)

Q1. Explain the role of feedback in the communication cycle.

Feedback is the receiver's response to the sender's message. It is the final component of the communication cycle. Without feedback, the sender cannot confirm whether the message was received, decoded correctly, or misunderstood, making the process incomplete.

Q2. Differentiate between interpersonal and physical barriers to communication with examples.

Physical barriers are environmental factors (e.g., loud machinery noise, long distance, poor Wi-Fi connectivity) that block the travel of a message. Interpersonal barriers are psychological states (e.g., personal prejudice, extreme anger, or lack of trust between the sender and receiver) that distort understanding.

Long Answer Questions (5 Marks)

Q1. What are the 7 Cs of effective communication? Explain any four in detail with examples of poor and good implementation.

The 7 Cs are Clear, Concise, Concrete, Correct, Coherent, Complete, and Courteous.

1. **Clear:** The message must be easy to comprehend. *Poor:* "I need the AI work done soon." *Good:* "Please submit Python Program 1 to 5 by tomorrow at 2 PM."
2. **Concise:** Keep it short. *Poor:* "In order to complete our assignment, I was thinking we should write the code which is given in Chapter 5 in the book today." *Good:* "Let's write Chapter 5's code today."
3. **Complete:** Provide all details. *Poor:* "Meeting tomorrow." *Good:* "AI group meeting tomorrow in Lab 2 at 11:30 AM to discuss SDGs."
4. **Courteous:** Respectful tone. *Poor:* "Give me your notebook." *Good:* "Could you please share your notes with me for reference?"

Self-Management Skills-I

Self-management is the ability to guide your own thoughts, emotions, and behaviors productively. This unit teaches self-awareness, identifying strengths and weaknesses, cultivating self-confidence, harnessing positive thinking, and practicing personal hygiene and grooming.

2.1 Introduction & Strengths/Weaknesses Analysis (SWOC)

Self-management helps us take control of our lives, build resilience, and work towards goals without needing constant supervision. It begins with **self-awareness**—knowing who you are, what you are capable of, and what you need to improve.

Strength and Weakness Analysis

An excellent tool for self-awareness is the **SWOC Analysis** (Strengths, Weaknesses, Opportunities, Challenges/Threats). For class 9 students, this focus is primary on identifying internal traits:

STRENGTHS (INTERNAL)

- Things you do exceptionally well.
- Skills, talents, and knowledge.
- Positive habits and attitude.
- *Example:* "I am good at math and Python logic."

WEAKNESSES (INTERNAL)

- Areas where you struggle or need help.
- Negative habits or lack of skills.
- Limiting beliefs.
- *Example:* "I get nervous speaking in front of the class."

SWOC Analysis Framework

S - Strengths

What am I good at?
Skills & talents

INTERNAL (+)

W - Weaknesses

Where do I struggle?
Areas to improve

INTERNAL (-)

O - Opportunities

What chances exist?
Courses, mentors

EXTERNAL (+)

C - Challenges

What obstacles exist?
Competition, fear

EXTERNAL (-)

Internal Factors (Within You)

External Factors (Outside You)

Activity: Fill in your own SWOC chart. Write 3 items under each category.

This self-reflection helps you set realistic goals and plan your growth areas.

FIGURE A2.1: SWOC ANALYSIS — SELF-AWARENESS FRAMEWORK

How to Identify Strengths & Weaknesses

To identify your strengths, ask yourself: What do I enjoy doing? What do others praise me for? What tasks do I complete easily? To identify your weaknesses: What tasks do I avoid? Where do I face constant failure? What feedback do teachers or parents give me for improvement?

2.2 Self-Confidence and Positive Thinking

Self-confidence is trust in one's own abilities, qualities, and judgment. It is not an innate trait; it can be developed over time.

FACTORS INFLUENCING SELF-CONFIDENCE

- **Social Factors:** Positive reinforcement from family, friends, and teachers boosts confidence. Bullying or constant comparison lowers it.
- **Cultural Factors:** Cultural beliefs, family expectations, and community standards can form a student's self-image.
- **Physical/Personal Factors:** Physical health, self-talk (inner dialogue), and previous experiences of success or failure.

Positive Thinking

Positive thinking is a mental attitude where you expect good and favorable results. It does not mean ignoring problems, but rather facing them with a constructive approach.

- **Benefits:** Reduces stress, improves physical and mental health, increases focus, and builds problem-solving skills.
- **Techniques to Build Positive Thinking:** Use positive affirmations, visualize success, learn from failures, surround yourself with supportive people, and practice gratitude.

2.3 Personal Hygiene and Grooming

How we look and care for our bodies directly impacts our self-esteem, how others view us, and our physical health.

Personal Hygiene

Practicing good personal hygiene prevents the spread of diseases and ensures we feel fresh and confident:

- **Skin Hygiene:** Bathing daily to wash away sweat, dirt, and dead cells.
- **Hand Hygiene:** Washing hands with soap before meals, after visiting the restroom, and after touching dirty surfaces.
- **Oral Hygiene:** Brushing teeth twice a day (morning and night) and rinsing after meals.
- **Hair Hygiene:** Washing and combing hair regularly to prevent dandruff and infections.

Grooming

Grooming is the process of keeping oneself neat, tidy, and appropriately dressed.

ASPECT	GOOD GROOMING HABITS	IMPACT ON SELF-IMAGE
Clothing	Clean, ironed clothes or school uniform. Shoes polished. No stains.	Creates a professional impression, showing respect to others.
Hair & Nails	Hair trimmed and neatly combed. Nails cut short and kept clean.	Prevents bacterial storage under nails; shows orderliness.
Body Odor	Use of mild deodorants/talcum powder. Regular washing of sweaty areas.	Prevents discomfort to peers; boosts social confidence.

UNIT SUMMARY

Self-management equips us to analyze our internal traits through SWOC, cultivate self-confidence by managing social/cultural factors, maintain positive thinking in difficult situations, and maintain personal hygiene/grooming for physical wellness and self-respect.

Unit 2 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. Which of the following is an internal factor in SWOC Analysis?

- a) Opportunities
- b) Weaknesses
- c) Threats
- d) Competitor growth

Answer: b

Q2. Maintaining clean, cut fingernails is an example of:

- a) Visual Communication
- b) Personal Hygiene and Grooming
- c) Strength Analysis
- d) Interpersonal dynamics

Answer: b

Short Answer Questions (2-3 Marks)

Q1. Explain the difference between a strength and a weakness. Give one example of each.

A strength is an internal quality or skill that you are good at (e.g., having a logical mind for coding). A weakness is an internal area where you struggle and need improvement (e.g., procrastinating and submitting assignments late).

Q2. List three ways to build self-confidence.

Three ways to build self-confidence are: (1) Setting realistic goals and achieving them, (2) Practicing positive self-talk and avoiding self-doubt, and (3) Learning from past mistakes rather than dwelling on them.

Information & Communication Technology (ICT) Skills-I

ICT skills are critical in today's digital era. This unit introduces basic computer hardware, operating systems, file management, the role of internet and email, and cyber safety rules.

3.1 Basics of ICT, Hardware and Software

Information and Communication Technology (ICT) refers to all forms of technology used to create, store, process, display, and exchange information in electronic format.

Hardware vs. Software

A computer system consists of two major components:

- **Hardware:** The physical, tangible components of a computer that you can see and touch. Examples: CPU, RAM, Hard Disk, Keyboard, Monitor, Motherboard.
- **Software:** A set of instructions or programs that tells the hardware what to do. It cannot be touched. Examples: Operating systems, MS Office, web browsers, games.

Types of Software

CATEGORY	DESCRIPTION	EXAMPLES
System Software	Controls the basic operations of the computer and manages hardware resources.	Operating Systems (Windows, macOS, Linux, Android), device drivers.
Application Software	Programs designed to perform specific tasks for the user.	MS Word, Google Chrome, VLC Player, Python IDLE, Tableau.

3.2 Operating System & File Management

An **Operating System (OS)** acts as an interface between the user and the computer hardware.

KEY FUNCTIONS OF AN OS

- **Processor Management:** Allocates CPU time to different tasks.
- **Memory Management:** Allocates and de-allocates RAM for running programs.
- **File Management:** Tracks where files are stored on the hard drive.
- **Device Management:** Communicates with printers, keyboards, and mouse via drivers.

File Management Basics

All data in a computer is stored in files. A **file** is a collection of related information. A **folder (directory)** is a container used to organize files.

- **File Extensions:** A suffix at the end of a filename indicating its format (e.g., .txt for text, .html for web page, .py for Python code, .pdf for document).
- **Paths:** The address of a file (e.g., C:\Book\Class 9th AI\Class-9-AI-Premium-Textbook.html).

3.3 Internet, Email, and Cyber Safety

The **Internet** is a global network of interconnected computers communicating via standardized protocols (TCP/IP).

Key Internet Terms

- **World Wide Web (WWW):** A collection of multimedia web pages accessible via the internet.

- **Web Browser:** Software application used to access the WWW (e.g., Google Chrome, Firefox, Edge).
- **Search Engine:** A tool used to search information on the web (e.g., Google, Bing).
- **Email:** Electronic Mail used to send letters, text, and attachments instantly across networks.

ESSENTIAL CYBER SAFETY RULES

Using the internet carries risks (viruses, identity theft, cyberbullying). Practice these safety tips:

1. **Strong Passwords:** Create complex passwords mixing uppercase, lowercase, numbers, and symbols. Avoid common words or birth dates.
2. **Do Not Share Personal Info:** Never post address, phone numbers, or passwords publicly.
3. **Phishing Awareness:** Do not click on links or download attachments from unknown senders.
4. **Anti-Virus:** Keep an active, updated antivirus software running on your computer.
5. **Safe Browsing:** Only visit sites starting with `https://` (the 's' stands for secure).

Unit 3 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. Which of the following is system software?

- a) MS PowerPoint
- b) Windows 11
- c) Google Chrome
- d) Python IDLE

Answer: b

Q2. What is the file extension used for Python scripts?

- a) .txt
- b) .html
- c) .py
- d) .docx

Answer: c

Short Answer Questions (2-3 Marks)

Q1. What is the purpose of an operating system? List two examples.

An operating system acts as a bridge between computer hardware and user applications. It manages computer resources like CPU, RAM, storage, and peripheral devices. Examples include Windows, Linux, macOS, and Android.

Q2. Define a file path and write a hypothetical path of a file named "python_intro.py" inside a folder "AI_Class9" in the D drive.

A file path is the complete address or route to a specific file within the storage hierarchy. The path for "python_intro.py" inside the "AI_Class9" folder on the D drive would be: `D:\AI_Class9\python_intro.py`.

Entrepreneurial Skills-I

Entrepreneurs drive innovation, economic growth, and societal change. This unit outlines what entrepreneurship is, explains the roles entrepreneurs play in our communities, describes their key traits, and breaks down the main types of business setups.

4.1 Concept of Entrepreneurship

An **entrepreneur** is a person who sets up a business or businesses, taking on financial risks in the hope of profit. The process of setting up, developing, and running a business is called **entrepreneurship**.

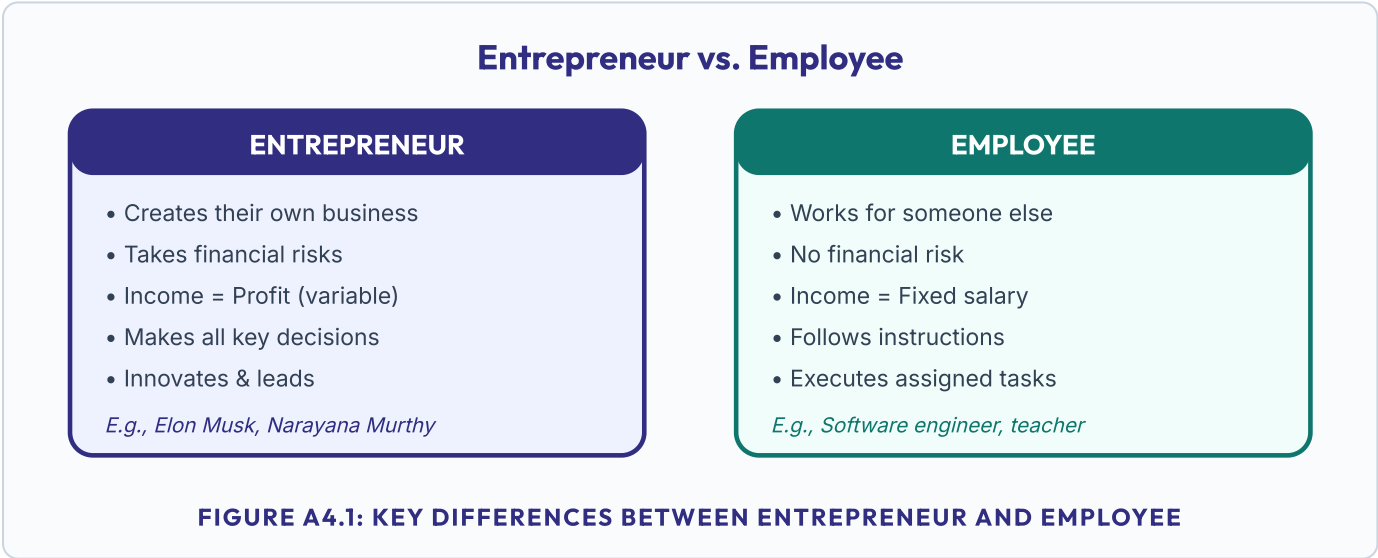
ENTREPRENEUR VS. EMPLOYEE

- **Entrepreneur:** An owner who takes risks, innovates, organizes resources, and works to create values and earn profit. They are self-employed.
- **Employee:** A worker hired by an employer to perform specific duties in exchange for a fixed salary or wage. They do not share the business risk.

Role of Entrepreneurs in Society

Entrepreneurs play a vital role in local and national development:

- **Wealth Generation:** Creating products and services that generate revenue and boost local markets.
- **Job Creation:** Hiring local workers, reducing unemployment, and improving community livelihood.
- **Innovation:** Finding creative solutions to everyday problems (e.g., smart home AI devices, digital payment portals).
- **Regional Development:** Establishing businesses in remote or developing areas, building local infrastructure.



4.2 Characteristics of a Successful Entrepreneur

Not every business owner succeeds. Successful entrepreneurs typically share these key characteristics:

PERSONAL TRAITS

- **Patience & Perseverance:** Remaining steady in the face of setbacks.
- **Confidence:** Believing in their ideas and abilities.
- **Hard Work & Dedication:** Willingness to work long hours to set up the firm.

BUSINESS TRAITS

- **Risk-Taking:** Willingness to invest capital without guaranteed returns.
- **Creativity & Innovation:** Designing unique products or finding better workflows.
- **Leadership & Teamwork:** Inspiring employees to work towards a shared vision.

KEY DISTINCTION

Running a Simple Business vs. Entrepreneurship: Opening a general store that sells standard grains is a simple business. Developing an app that connects local farmers directly to customers to sell grains using AI demand forecasting is entrepreneurship.

4.3 Types of Businesses

Businesses can be classified based on the nature of their activities:

BUSINESS TYPE	DESCRIPTION	EXAMPLES
Product-based	Sells tangible, physical items that customers can touch and take home.	Clothing store, bookstore, phone store, bakery.
Service-based	Sells intangible skills, labor, or expertise to solve a customer's problem.	Software company, tutoring center, hair salon, plumbing, cleaning service.
Hybrid	Combines selling both products and services in a single business.	A restaurant (sells food products + waiting services), a car dealer with a repair shop.

Unit 4 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. A restaurant is an example of which type of business?

- a) Service-based only
- b) Product-based only
- c) Hybrid
- d) Non-profit

Answer: c

Q2. What is the primary motive of an entrepreneur?

- a) To work fixed hours
- b) To avoid all business risks
- c) To innovate, create values, and build a profitable business
- d) To receive a fixed monthly salary

Answer: c

Short Answer Questions (2-3 Marks)

Q1. State two roles of entrepreneurs in a society.

Two major roles of entrepreneurs are: (1) Creating job opportunities for local citizens, which reduces unemployment, and (2) Introducing innovative products or services that solve specific problems and improve quality of life.

Q2. Explain risk-taking as a trait of an entrepreneur.

Risk-taking means the willingness of an entrepreneur to invest their time, money, and effort into a new business venture despite the uncertainty of whether the business will yield a profit or result in financial loss.

Green Skills-I

A healthy ecosystem is essential for human survival and economic progress. This unit explores ecosystems, how human activities affect natural systems, the goals of sustainable development, and actions individuals can take to build a green economy.

5.1 Ecosystems, Society, and Environment

An **ecosystem** is a community of living organisms (biotic factors) interacting with their non-living environment (abiotic factors) like air, water, soil, and sunlight.

BIOTIC VS. ABIOTIC COMPONENTS

- **Biotic Factors:** Plants, animals, fungi, bacteria, and humans.
- **Abiotic Factors:** Water, temperature, wind, sunlight, minerals, and soil.

Human Impact on the Environment

Rapid industrialization, urban growth, and high resource consumption have disrupted ecosystems:

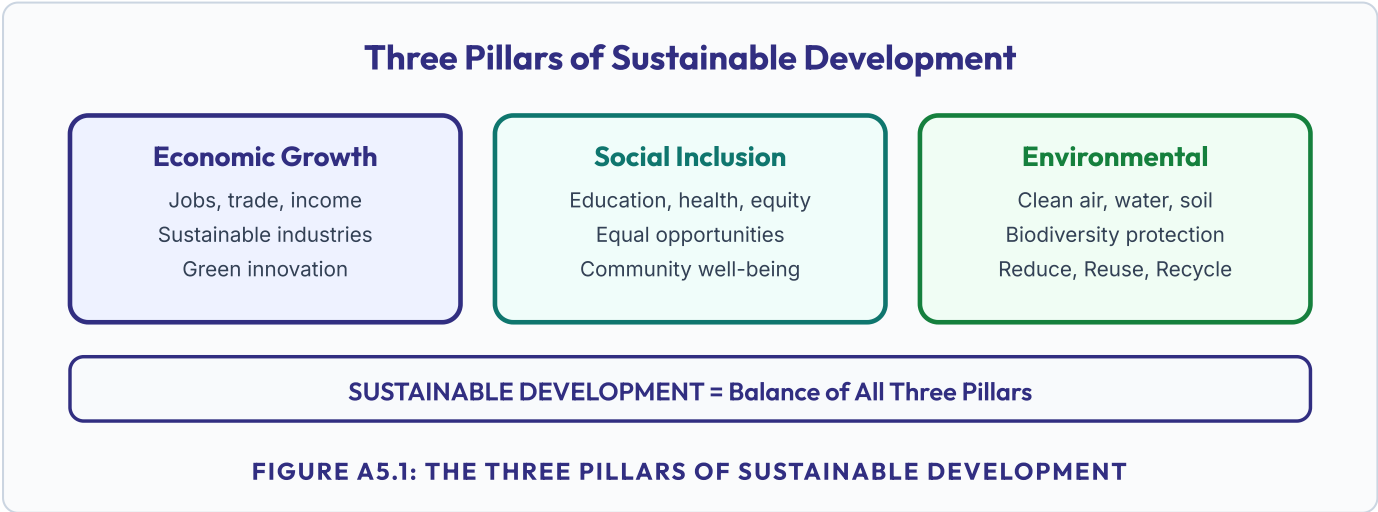
- **Pollution:** Air (caused by factories/vehicles), Water (from industrial waste/sewage), and Soil pollution (from pesticides and plastic dumping).
- **Deforestation:** Cutting down forests for agriculture or urban spaces, leading to soil erosion and habitat loss.
- **Global Warming:** Excessive emission of greenhouse gases (carbon dioxide, methane) trapping heat in the atmosphere, leading to climate change.

5.2 Sustainable Development & The Green Economy

To balance economic development with environmental protection, we must practice **sustainable development**.

DEFINITION OF SUSTAINABLE DEVELOPMENT

Sustainable Development is development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs. It rests on three pillars: Economic growth, Social inclusion, and Environmental protection.



The Green Economy

A **green economy** is an economy that aims for sustainable development without degrading the environment. It is low-carbon, resource-efficient, and socially inclusive.

Individual Role: The 3 Rs and Actions

Every student and citizen can contribute to environmental conservation through the **3 Rs**:

- **Reduce:** Limit the use of non-biodegradable or resource-intensive materials (e.g., carry reusable cloth bags, switch off idle lights).
 - **Reuse:** Find new ways to use items instead of throwing them away (e.g., reuse plastic bottles as planting pots).
 - **Recycle:** Process waste materials into new products (e.g., segregate wet and dry waste so paper, glass, and metal can be recycled).
-

Unit 5 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. Which of the following is an abiotic component of an ecosystem?

- a) Soil bacteria
- b) Decayed leaves
- c) Temperature and sunlight
- d) Fungi

Answer: c

Q2. Meeting our current needs without harming the resources of future generations is called:

- a) Industrialization
- b) Sustainable Development
- c) Green economy shift
- d) Deforestation balance

Answer: b

Short Answer Questions (2-3 Marks)

Q1. Define sustainable development and list its three pillars.

Sustainable development is a development model that satisfies present needs while preserving natural systems and resources for the future. Its three pillars are: (1) Environmental protection, (2) Social equality, and (3) Economic development.

Q2. Explain how the 'Reduce' principle helps the environment, with an example.

The 'Reduce' principle means using fewer natural resources and generating less waste. For example, turning off lights and fans when leaving a room reduces electricity consumption, which decreases the amount of coal burned in thermal power plants.

Long Answer Questions (5 Marks)

Q1. What is a Green Economy? Discuss three ways in which citizens can help build a green economy.

A Green Economy is one that promotes sustainability, reduces environmental risks, and ensures ecological balance. It focuses on reducing carbon footprint and using clean energy sources.

Citizens can help in three ways:

1. **Energy Conservation:** Using energy-efficient appliances (LEDs), turning off idle electronics, and utilizing solar energy.
2. **Waste Segregation and Recycling:** Separating wet (biodegradable) and dry (recyclable) waste at home, and minimizing the use of single-use plastics.
3. **Sustainable Transportation:** Walking, cycling, using public transport, or carpooling to reduce fuel consumption and greenhouse gas emissions.

AI Reflection, Project Cycle and Ethics

Artificial Intelligence is reshaping our relationship with technology. This unit reflects on what AI is, explores its three core domains (Data, Computer Vision, NLP), breaks down the six stages of the AI Project Cycle, and examines the ethical issues of bias, access, and decision-making.

1.1 AI Reflection & Three Realms of AI

We interact with Artificial Intelligence daily, often without realizing it. From Google's search predictions and YouTube recommendations to smart voice assistants and face unlock features on smartphones, AI is embedded in modern life.

WHAT IS ARTIFICIAL INTELLIGENCE?

Artificial Intelligence (AI) is a branch of computer science that deals with building smart machines capable of performing tasks that typically require human intelligence—such as learning, reasoning, problem-solving, perception, and understanding natural language.

The Three Domains of AI

AI applications are categorized into three core domains based on the type of data they process and the capabilities they enable:

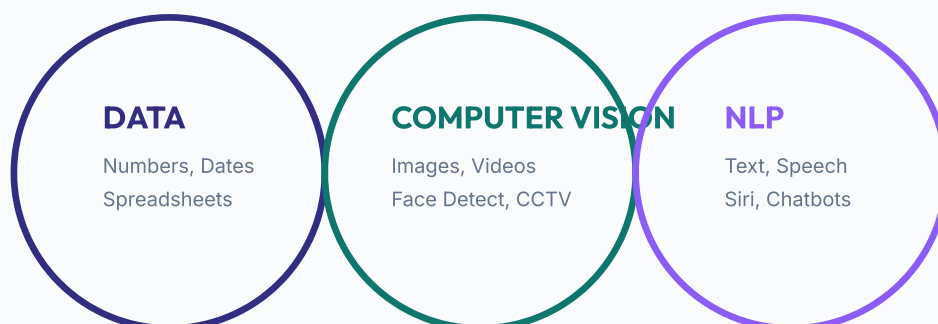


FIGURE 2.1: THE THREE DOMAINS OF ARTIFICIAL INTELLIGENCE

- 1. Data Sciences (Data-based AI):** Processes alphanumeric data (numbers, tables, spreadsheets). It is used for prediction, trend analysis, and pattern matching.
Gamified Reference: Rock, Paper, Scissors (learns patterns from your history of throws).
- 2. Computer Vision (CV):** Enables machines to recognize, process, and interpret visual data (images and videos) in the same way humans do.
Gamified Reference: Google's Quick Draw (classifies doodles using neural networks).
- 3. Natural Language Processing (NLP):** Focuses on enabling computers to understand, interpret, and generate human languages (spoken or written).
Gamified Reference: Google's Semantris (associates words semantically).

SMART HOMES: AN INTEGRATION EXAMPLE

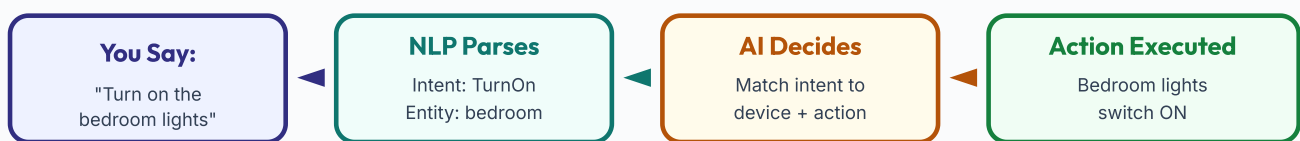
A smart home integrates all three domains:

- **NLP:** Voice assistant (Alexa/Google Assistant) interprets a statement: "Turn on the living room lights."
- **Data:** Smarthome logs temperature, occupancy data, and power usage to optimize air conditioning.
- **Computer Vision:** Smart security cameras monitor the gate, performing facial recognition to unlock doors for family members.

RECOMMENDED ACTIVITY: LUIS SMART HOME DEMO

Try Microsoft's LUIS (Language Understanding Intelligent Service) demo at <https://aidemos.microsoft.com/luis/demo>. Make a statement about lighting (e.g., "Turn the bedroom lights to 50%") and watch how the AI interprets your natural language command and adjusts the virtual house accordingly. This demonstrates how NLP powers real smart-home automation.

How LUIS Understands Your Command



This is how voice assistants like Alexa, Google Home, and Siri understand and act on your commands using NLP.

FIGURE 1.2: HOW NLP POWERS SMART HOME VOICE COMMANDS (LUIS DEMO)

1.2 The AI Project Cycle

Developing an AI solution is an iterative process. The **AI Project Cycle** outlines the six phases of this workflow:

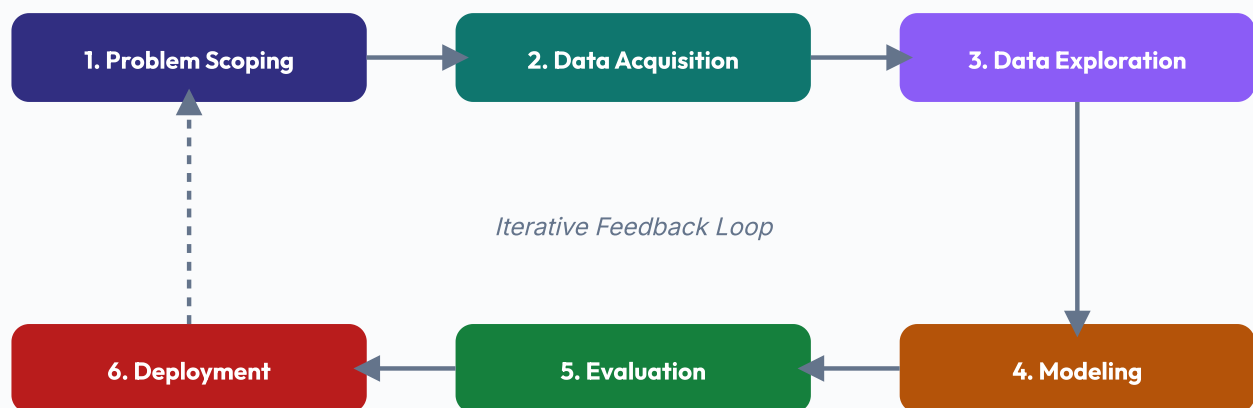


FIGURE 2.2: THE SIX PHASES OF THE AI PROJECT CYCLE

Phase 1: Problem Scoping

Before starting any project, we must identify and define the problem clearly. Setting goals requires identifying who is affected, what the parameters are, where it occurs, and why it is worth solving. To accomplish this, we use the **4Ws Problem Canvas**:

WHO?

Stakeholders affected by the issue.
E.g., Students, local patients.

WHAT?

Nature of the problem, evidence of its existence.
E.g., High rate of preventable blindness.

WHERE?

Context, location, and environment.
E.g., Rural clinics, government schools.

WHY?

Benefits of solving the problem; value generated
E.g., Reduces cost, improves quality of life.

FIGURE 2.3: THE 4WS PROBLEM CANVAS FRAMEWORK

PROBLEM STATEMENT TEMPLATE

Our **[Stakeholders]** in **[Where]** face the problem of **[What]**. A viable solution would **[Why - value proposition]**.

Activity: Problem Mind Map

A **mind map** is a visual brainstorming tool used to list down and organize problems related to a selected topic. Choose a theme (e.g., "Water Pollution in My City") and draw connections to sub-problems, then select one problem as your project goal.

Mind Map — Brainstorming Problems Around a Theme

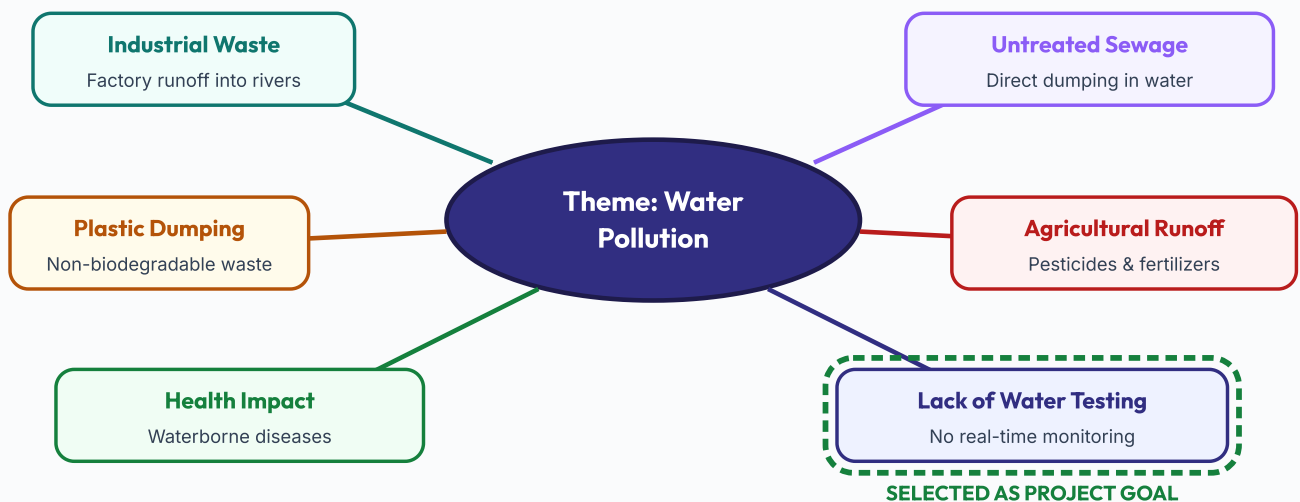


FIGURE 1.3: PROBLEM MIND MAP — SELECT ONE PROBLEM AS YOUR AI PROJECT GOAL

ACTIVITY: BUILD YOUR OWN MIND MAP

1. Choose a theme (e.g., "Traffic in My City", "Waste Management", "Student Health").
2. Brainstorm at least 5-6 sub-problems branching from the theme.
3. Select one problem as your AI project goal and fill in the 4Ws canvas for it.

Activity: Data and Analysis Planning

Before collecting data, you must plan what you need. Answer the following 8 guiding questions for your chosen problem:

8 Guiding Questions for Data & Analysis Planning

1

What are the data features needed?

2

How will the features collected affect the problem?

3

Where can you get the data?
(surveys, sensors, websites, APIs)

4

How frequently do you need to collect the data?

5

What happens if you don't have enough data?

6

What kind of analysis needs to be done?

7

How will the analysis be validated?

8

How does the analysis inform the action?

FIGURE 1.4: THE 8 GUIDING QUESTIONS FOR DATA & ANALYSIS PLANNING

Phase 2: Data Acquisition

Data is the raw fuel for AI. This phase focuses on gathering the necessary **data features** (attributes or parameters that impact our problem).

- **Sources of Data:** Surveys, sensors, databases, web scraping, government websites (e.g., data.gov.in), and APIs.
- **System Maps:** A visual representation designed to show how different data features interact with one another in a system (e.g., mapping how rainfall, soil moisture, and humidity affect crop yield).

TEAMWORK ACTIVITY: PRESENTATION – GOAL, ACTIONS & DATA

After completing your data acquisition plan, each team presents their work to the class:

1. **Present your Goal:** Share the problem statement and 4Ws canvas.
2. **Present your Actions:** Explain the stakeholders involved, current actions taken, and ethics considered.
3. **Present your Data Plan:** Show your system map and data features identified.
4. **Brainstorm Solutions:** As a class, brainstorm possible AI-enabled solutions for each team's problem statement. Discuss which approaches seem most feasible and why.

This teamwork activity builds communication skills and helps validate your approach through peer feedback.

Phase 3: Data Exploration

To understand acquired data, we must visualize it. Visualization helps us spot patterns, outliers, anomalies, and relationships.

- **Common Visualization Tools:** Bar graphs (comparisons), Line graphs (trends over time), Pie charts (proportions), Scatter plots (relationship between variables), and spreadsheets.
- **Explore Visualizations Online:** Visit <https://datavizcatalogue.com/> to discover different types of charts and when to use them.

ACTIVITY: TOP 10 SONG PREDICTION

Can you predict which songs will be in the Top 10 next month? Follow these steps:

1. **Identify Data Features:** What makes a song popular? (genre, artist followers, beats per minute, release season, social media mentions, etc.)
2. **Collect Data:** Record data for 15-20 current songs in a spreadsheet (use music charts and streaming sites).
3. **Visualize:** Create bar charts comparing features of Top 10 vs. non-Top 10 songs.
4. **Analyze:** Can you spot patterns? Do popular songs share common features?

This activity practices the full Data Exploration cycle: feature identification → collection → graphical representation → pattern discovery.

Data Visualization Types — When to Use Each

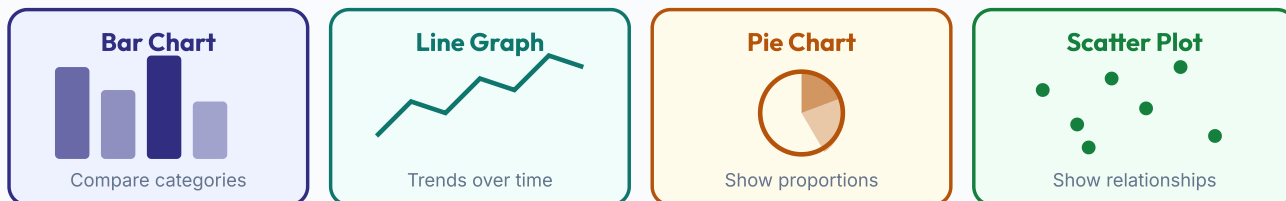


FIGURE 1.6: WHEN TO USE EACH VISUALIZATION TYPE — QUICK REFERENCE

Phase 4: Modeling

Modeling is the design stage of the actual AI brain. We write logic to process data and extract conclusions. There are two primary types of models:

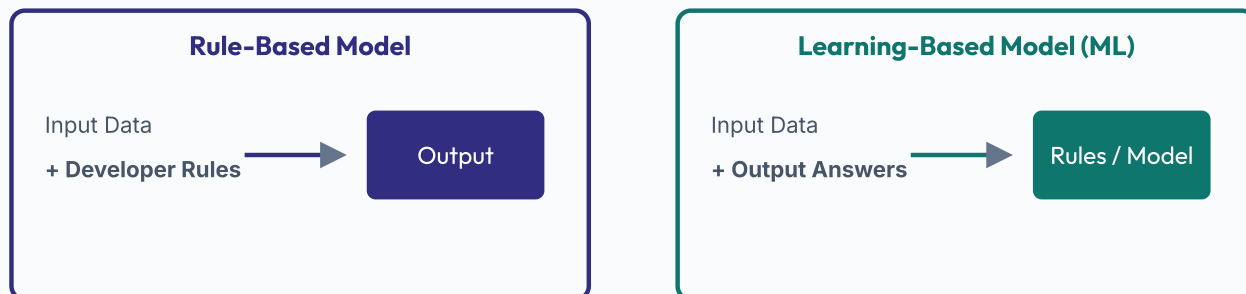


FIGURE 2.4: RULE-BASED VS. LEARNING-BASED WORKFLOWS

- **Rule-Based Modeling:** The developer writes explicit, hard-coded rules (e.g., if temperature > 35: state = 'hot'). The machine acts strictly on these instructions.
- **Learning-Based Modeling:** The machine is trained on data (inputs and outputs) and automatically extracts the underlying rules or patterns using mathematical algorithms (Machine Learning).

Phase 5: Evaluation

Once modeled, the AI must be tested with unseen data to check its accuracy. We evaluate it by constructing a **Confusion Matrix**, which tracks four cases:

Actual Status	Predicted: Yes	Predicted: No
Actual: Yes	True Positive (TP)	False Negative (FN)
Actual: No	False Positive (FP)	True Negative (TN)

FIGURE 2.5: THE 2X2 EVALUATION CONFUSION MATRIX

- **True Positive (TP):** The event is positive, and the AI correctly predicts it as positive (e.g., a patient has a disease, and the model flags it).
- **False Positive (FP):** The event is negative, but the AI incorrectly predicts it as positive (e.g., a healthy person is flagged as diseased—a false alarm).
- **False Negative (FN):** The event is positive, but the AI incorrectly predicts it as negative (e.g., a diseased patient is declared healthy—a dangerous miss).
- **True Negative (TN):** The event is negative, and the AI correctly predicts it as negative (e.g., a healthy person is declared healthy).

Phase 6: Deployment

The final stage where the model is integrated into a real-world system so users can benefit from it.

DEPLOYMENT CASE STUDY: PREVENTABLE BLINDNESS

Diabetic Retinopathy causes blindness but is treatable if detected early. There is a severe shortage of eye specialists in rural India. By deploying a deep learning model on cloud servers accessible via simple smartphone cameras in village clinics, medical staff can scan patients' eyes and instantly receive a referral recommendation if signs of retinopathy are detected, successfully preventing blindness.

ACTIVITY: AI MODEL FOR PERSONALIZED EDUCATION

Apply the complete AI Project Cycle to design an AI model that personalizes learning for students:

1. **Problem Scoping:** Students learn at different speeds. One-size-fits-all teaching leaves fast learners bored and slow learners behind.
2. **Data Acquisition:** Collect data features like quiz scores, time spent per topic, difficulty ratings, and learning style preference (visual/auditory/reading).
3. **Data Exploration:** Visualize which topics take longest, which students score low on similar question types.
4. **Modeling:** A learning-based model that recommends easier practice problems if a student scored low, and harder challenges if they scored high.
5. **Evaluation:** Test the model with a group of students — did their scores improve? Check TP/FP/TN/FN for correct recommendations.
6. **Deployment:** Integrate the model into a classroom learning app that adapts in real-time.

This activity challenges you to think about how deployment works in practice and how different deployment methods (app, website, chatbot) affect the user experience.

Personalized Education — AI Adapts to Each Student

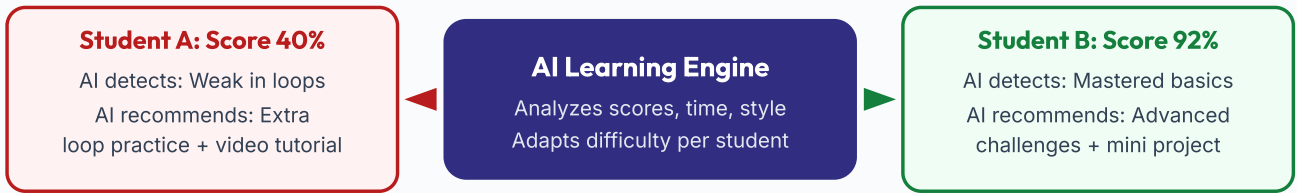


FIGURE 1.7: HOW AN AI PERSONALIZED EDUCATION MODEL ADAPTS TO DIFFERENT STUDENTS

1.3 Artificial Intelligence Ethics

As AI gains decision-making power, we must consider its ethical implications.

- **AI Bias:** AI models learn from historical data. If the historical data contains human biases (e.g., gender bias in hiring practices, racial bias in facial recognition), the AI will replicate and amplify these biases.
- **AI Access:** The digital divide. Will advanced AI tools be accessible to everyone, or will they only benefit wealthy countries and urban populations, leaving rural communities further behind?

ETHICS STUDY: THE MORAL MACHINE

Created by MIT, the Moral Machine (<https://www.moralmachine.net/>) explores the ethics of self-driving cars. In situations where an accident is unavoidable, how should the car's AI prioritize lives? Should it save passengers or pedestrians? Children or senior citizens? This demonstrates that AI logic requires moral and ethical considerations, not just numerical rules.

Activity: Ethics Awareness — Stakeholder Role-Play

In this activity, students play the role of major stakeholders in an AI scenario and decide what is ethical and what is not.

Ethics Awareness Role-Play — Facial Recognition in Schools

Scenario: A school wants to install AI facial recognition cameras
for attendance tracking and security. Is this ethical?

School Admin

"It saves time and improves security."

Likely: FOR

Student

"I feel watched all the time. No privacy."

Likely: AGAINST

Parent

"My child's face data could be misused."

CONCERNED

AI Company

"Our tech is accurate and data is encrypted."

Likely: FOR

How to Play

1. Divide class into groups — each group plays ONE stakeholder role.
2. Each group discusses and prepares 3-4 arguments from their stakeholder's perspective.
3. Hold a class debate. After all sides present, vote: Should the school install AI cameras? Discuss the final verdict.

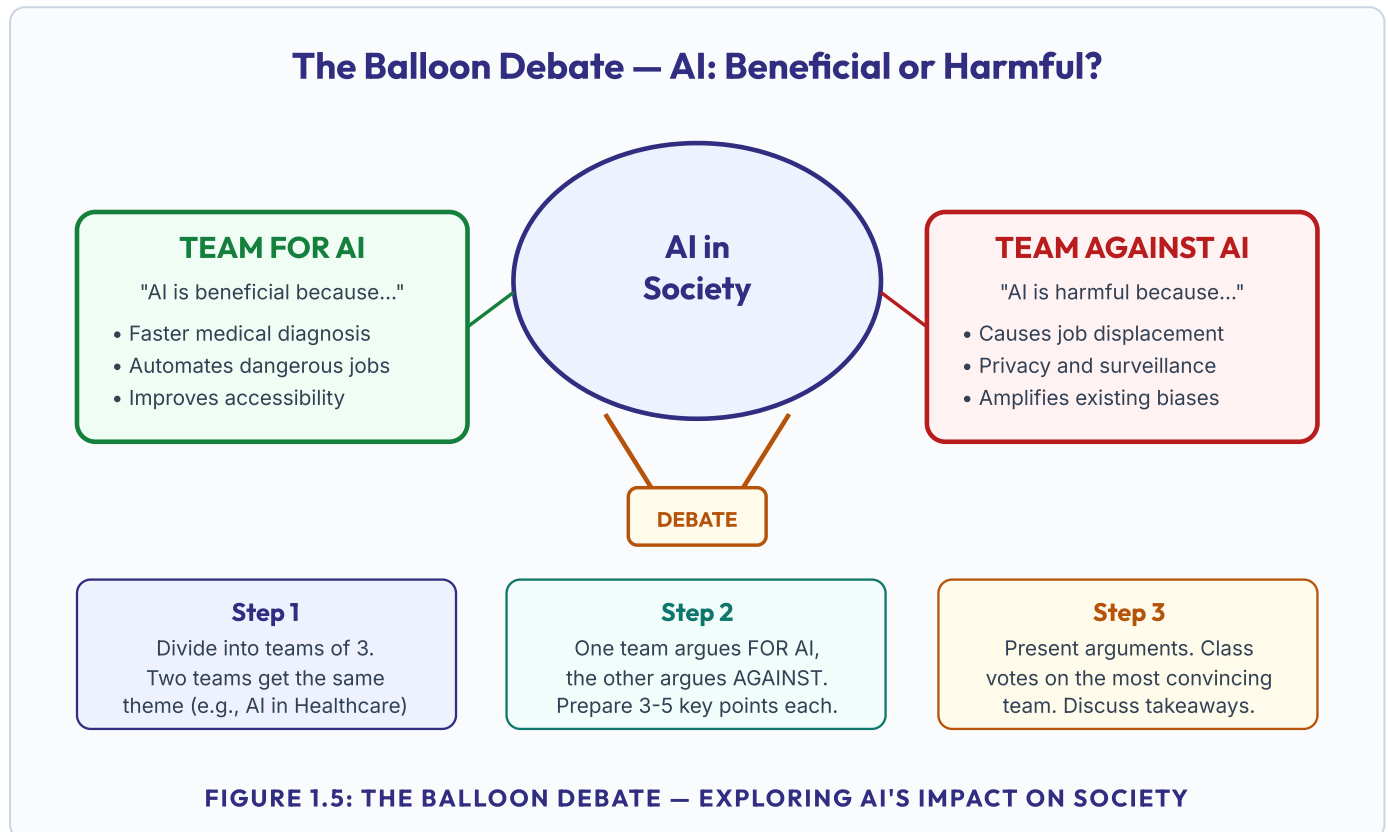
FIGURE 1.8: ETHICS AWARENESS — STAKEHOLDER ROLE-PLAY ACTIVITY

MORE SCENARIOS FOR ETHICS ROLE-PLAY

- **AI in Hiring:** A company uses AI to screen resumes. The AI rejects more female candidates. Who is responsible?
- **Self-Driving Cars:** An autonomous car must choose between hitting a pedestrian or swerving into a wall. Who decides the algorithm?
- **AI Health Diagnosis:** An AI diagnoses a patient incorrectly. Should the doctor, the AI company, or the hospital be held accountable?

Activity: The Balloon Debate

The Balloon Debate is a fun classroom activity to explore the pros and cons of AI in society. It encourages critical thinking and public speaking.



SAMPLE THEMES FOR BALLOON DEBATE

- **AI in Healthcare:** Should AI make medical diagnoses?
- **AI in Education:** Should AI replace human teachers?
- **AI in Employment:** Does AI create more jobs than it destroys?
- **AI in Creative Arts:** Can AI-generated art be considered real art?

Unit 1 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. In the 4Ws problem canvas, which canvas element identifies the target audience or users affected by the problem?

- a) What
- b) Who
- c) Where
- d) Why

Answer: b

Q2. An AI model incorrectly predicts that a healthy person has COVID-19. What evaluation case is this?

- a) True Positive
- b) False Negative
- c) False Positive
- d) True Negative

Answer: c

Short Answer Questions (2-3 Marks)

Q1. Explain the main difference between rule-based and learning-based modeling.

In rule-based modeling, developers write explicit logic rules for the system to process inputs and produce outputs. In learning-based modeling, mathematical models are trained on datasets containing both inputs and outputs, allowing the algorithm to automatically learn rules and correlations.

Q2. What is AI Bias? How does it sneak into an AI system?

AI Bias occurs when an AI system produces systematically prejudiced outputs. It enters the system through biased training datasets. If the data collected represents historic human discrimination, the model learns these patterns and replicates them.

Data Literacy

Data is the currency of the information age. This unit defines data literacy, details the Data Literacy Process Framework, contrasts data privacy with security, and introduces tools like Tableau for data dashboard visualization.

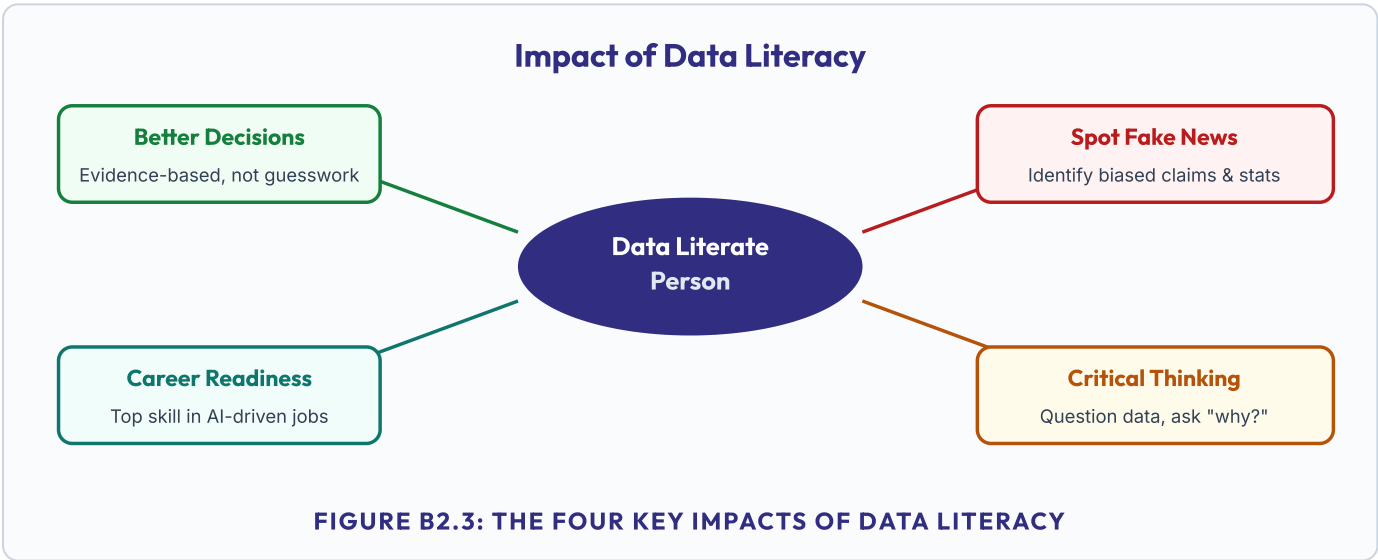
2.1 Basics of Data Literacy

With the rise of internet usage, vast amounts of data are generated every second. To navigate this world, we must develop data literacy.

DATA LITERACY DEFINED
Data Literacy is the ability to read, understand, analyze, work with, and argue with data. It empowers individuals to make data-driven decisions and identify false claims or biased reports in news articles and advertisements.

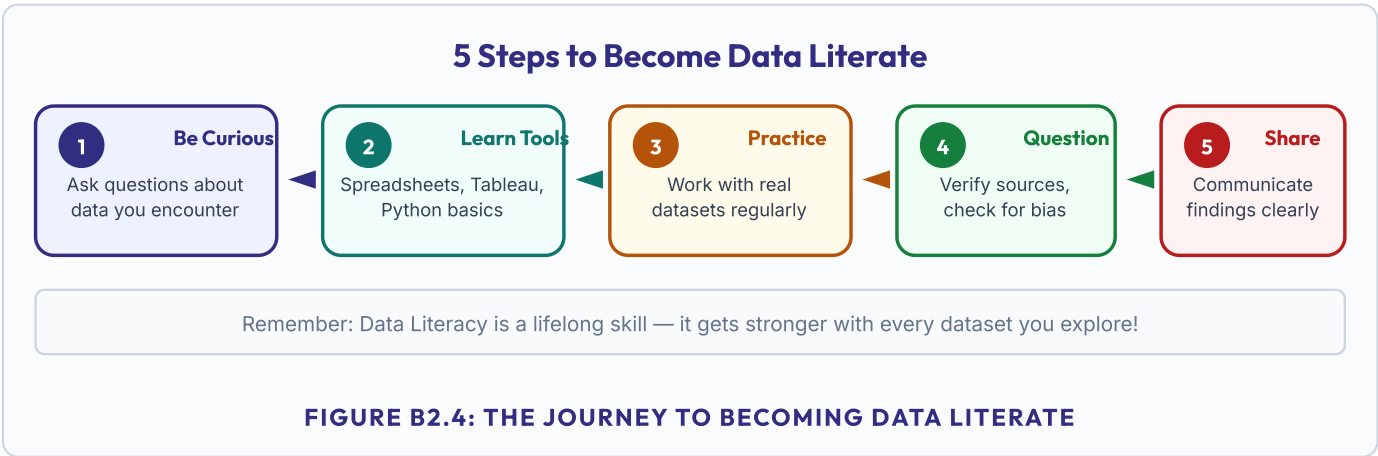
Impact of Data Literacy

Data literacy has a powerful impact across every area of modern life. Individuals who are data literate can make better personal, academic, and professional decisions.



How to Become Data Literate?

Becoming data literate is a journey, not a one-day task. Follow these steps to build your skills:



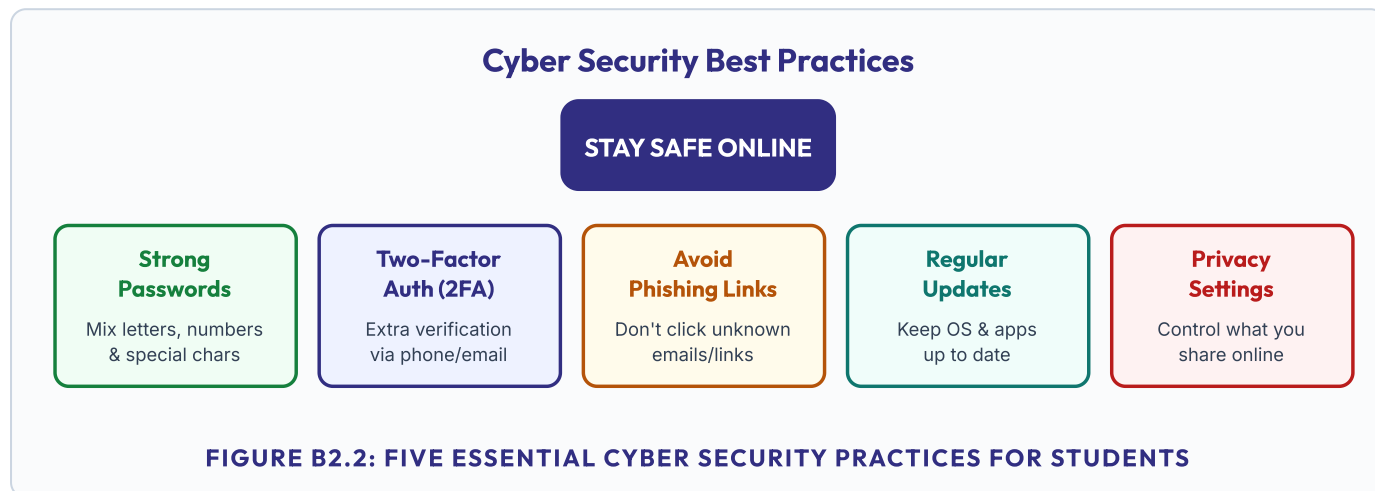
ACTIVITY: IMPACT OF NEWS ARTICLES

Find a news article that uses data or statistics (e.g., "80% of students prefer online learning").

1. **Read:** What claim does the article make? What data supports it?
2. **Question:** Who collected the data? How large was the sample? Could the data be biased?
3. **Verify:** Can you find the original source or study? Does the article misrepresent the data?
4. **Present:** Share your findings with the class. Was the news article data-literate or misleading?

This activity builds critical thinking — a core component of data literacy.

What are Data Security and Privacy? How Are They Related to AI?



Data Privacy vs. Data Security

Although often used interchangeably, data privacy and security refer to distinct concepts:

	DATA PRIVACY	DATA SECURITY
Definition	Deals with the proper handling of user data based on consent and regulations. It is about who has legal access to it.	Deals with protecting data from unauthorized access, theft, hacking, or malicious breaches.
Focus	User rights, transparency, and data sharing policies.	Encryption, firewalls, passwords, and physical hardware locks.
Analogy	Deciding which guests are invited to enter your home.	Installing lock systems and security cameras on your doors to keep thieves out.

HOW DATA PRIVACY & SECURITY RELATE TO AI

AI systems require **massive amounts of data** to train effectively. This creates unique challenges:

- **Privacy Risk:** AI models trained on personal data (photos, health records, browsing habits) can expose sensitive information if not properly anonymized.
- **Security Risk:** AI databases storing millions of user records become prime targets for hackers. A data breach can expose personal data of millions in seconds.
- **Bias from Data:** If private data is collected without diverse representation, the AI model inherits and amplifies societal biases.

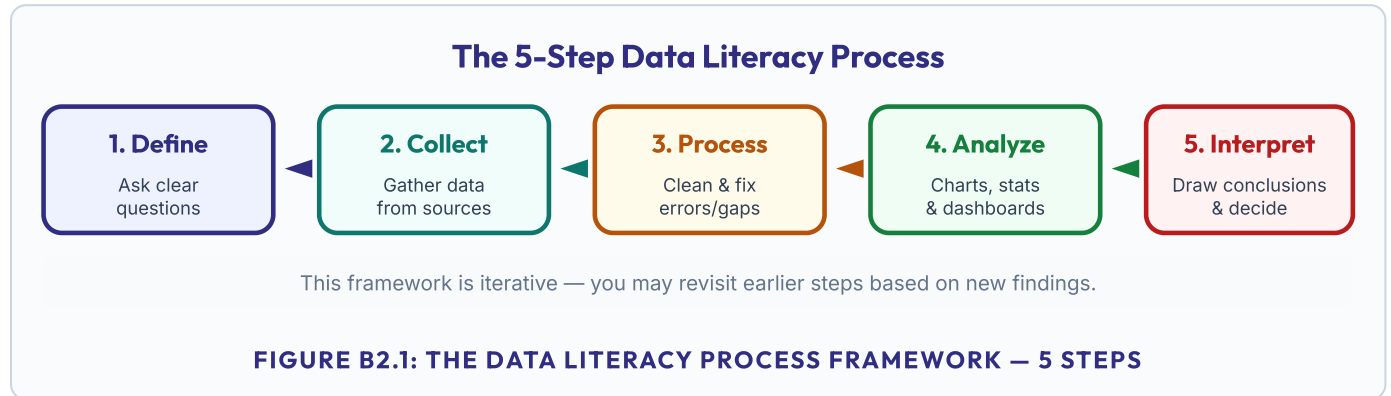
This is why data privacy laws (like GDPR) and strong security measures are essential when building AI systems.

REFERENCE RESOURCES

- **Video — What is Data Literacy?** https://www.youtube.com/watch?v=yhO_t-c3yJY
- **Video — Why Data Literacy Matters:** <https://www.youtube.com/watch?v=aO858HyFbKI>
- **CBSE Cyber Safety Guidelines (PDF):** <https://www.cbse.gov.in/cbsenew/documents/Cyber%20Safety.pdf>

2.2 The Data Literacy Process Framework

To work with data effectively, we follow a systematic process framework:



1. **Define:** Formulate clear, research-oriented questions that need answers.
2. **Collect:** Gather data from primary sources (interviews, experiments) or secondary sources (public repositories like WHO, Kaggle).
3. **Process & Clean:** Remove errors, empty values, duplicates, and irrelevant features (Data Preprocessing).
4. **Analyze & Visualize:** Use statistical tools, charts, and dashboards to find trends.
5. **Interpret:** Draw logical conclusions that help in decision-making.

DATA PREPROCESSING & FEATURE EXTRACTION

Raw data is often noisy and incomplete. Preprocessing steps include:

- **Handling Missing Data:** Deleting rows with empty values or filling them with mean values.
- **Data Formatting:** Ensuring dates, numbers, and strings are consistently structured.
- **Feature Selection:** Keeping only the columns that directly impact the output, which optimizes training speed.

2.2a Acquiring Data, Processing, and Interpreting Data

Types of Data

Before acquiring data, it is important to understand the different types of data we can work with:

Types of Data — A Classification

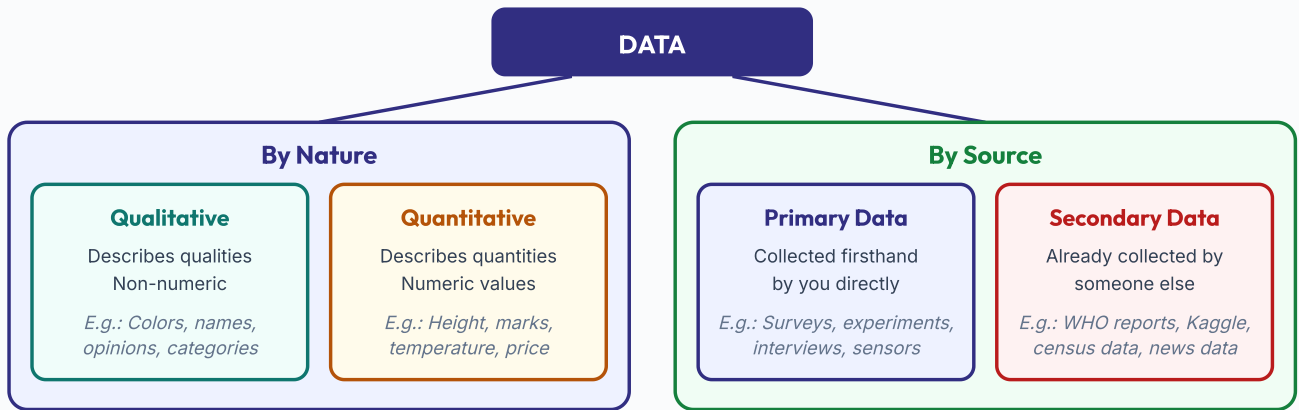


FIGURE B2.5: CLASSIFICATION OF DATA — BY NATURE AND BY SOURCE

Data Acquisition — Methods & Best Practices

Data Acquisition is the process of collecting relevant data for analysis. The method you choose depends on the problem, the type of data needed, and available resources.

Data Acquisition Methods



Best Practices: Ensure consent, verify reliability, use diverse sources, document collection methods, collect enough samples.

FIGURE B2.6: FIVE COMMON DATA ACQUISITION METHODS

Data Processing and Data Interpretation

After data is collected and preprocessed, the next critical step is **data interpretation** — the process of reviewing data, assigning meaning, and drawing conclusions from it.

DATA INTERPRETATION DEFINED

Data Interpretation is the process of making sense of collected data by analyzing patterns, relationships, and trends to arrive at meaningful conclusions that support decision-making.

Types of Data Interpretation

Two Types of Data Interpretation

Qualitative Interpretation

Analyzes non-numeric data
(opinions, themes, categories)

Methods:

- Content analysis (text/interviews)
- Coding & thematic grouping

Quantitative Interpretation

Analyzes numeric data
(numbers, statistics, measurements)

Methods:

- Statistical analysis (mean, mode)
- Graphical analysis (charts, trends)

FIGURE B2.7: QUALITATIVE VS QUANTITATIVE DATA INTERPRETATION

Importance of Data Interpretation

- **Informed Decision-Making:** Interpretation turns raw numbers into actionable insights for governments, businesses, and individuals.
- **Pattern Discovery:** Identifying trends that are invisible in raw data (e.g., seasonal spikes in sales or disease outbreaks).
- **Validating Hypotheses:** Testing whether assumptions are supported by evidence before acting on them.
- **Communicating Results:** Interpreted data is easier to communicate to stakeholders through visualizations and summary reports.

ACTIVITY: TREND ANALYSIS

Collect the daily maximum temperature of your city for the past 7 days from a weather website or newspaper. Record the data in a table:

DAY	MON	TUE	WED	THU	FRI	SAT	SUN
Temp (°C)	34	35	33	36	38	37	39

1. Plot a **line graph** with days on X-axis and temperature on Y-axis.
2. **Interpret:** Is there a rising or falling trend? What could cause the pattern?
3. **Predict:** Based on the trend, what might Monday's temperature be?

ACTIVITY: VISUALIZE AND INTERPRET DATA

Collect data on the favourite sports of 20 classmates. Record the results and:

1. Create a **bar chart** showing the frequency of each sport.
2. Create a **pie chart** showing the percentage distribution.
3. **Interpret:** Which sport is most popular? Which is least popular? What does the data tell you about your class?
4. **Present:** Share your visualizations and interpretations with the class.

Use spreadsheet software (Google Sheets or Excel) or Tableau to create your charts.

2.3 Data Dashboards & Visualization using Tableau

A **data dashboard** is an information management tool that visually tracks, analyzes, and displays key performance indicators (KPIs), metrics, and data points to monitor the health of a department or project.

Visualization Tools

Data Visualization Tools for Students

Tableau Public

Drag-and-drop dashboards
Interactive charts & filters

public.tableau.com/en-us/s/download

Datawrapper

Quick charts from pasted data
Used by newsrooms worldwide

www.datawrapper.de

Google Sheets / Excel

Spreadsheet-based charts
Best for beginners & class use

sheets.google.com

FIGURE B2.8: RECOMMENDED DATA VISUALIZATION TOOLS

Tableau Public — Step-by-Step Guide

Tableau is a popular, free data visualization software used to convert raw spreadsheets into interactive charts and dashboards.

CREATING A DASHBOARD IN TABLEAU (STEP-BY-STEP)

1. **Connect to Source:** Import your data (e.g., an Excel file or Google Sheet).
2. **Understand Dimensions & Measures:**
 - *Dimensions:* Qualitative fields (e.g., Category, Region, Date) that slice your data.
 - *Measures:* Quantitative numeric fields (e.g., Sales, Profit, Temperature) that can be aggregated.
3. **Drag and Drop:** Drag dimensions to columns and measures to rows to automatically generate bar charts or scatter plots.
4. **Build a Dashboard:** Combine multiple sheets into a single dashboard container, allowing interactive filtering.

VIDEO TUTORIALS & REFERENCE LINKS

- **Tableau for Beginners (Tutorial):** <https://www.youtube.com/watch?v=NLCzpPRCc7U>
- **Create Your First Dashboard:** https://www.youtube.com/watch?v=_M8BnosAD78
- **Download Tableau Public:** <https://public.tableau.com/en-us/s/download>
- **Try Datawrapper (no sign-up needed):** <https://www.datawrapper.de/>

Unit 2 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. Which of the following belongs to "Measures" in Tableau?

- a) Order Date
- b) Customer Name
- c) Sales Profit Value
- d) Product Category

Answer: c

Q2. Encrypting database files to prevent hackers from stealing customer files is an example of:

- a) Data Privacy
- b) Data Security
- c) Data Preprocessing
- d) Data Analysis

Answer: b

Short Answer Questions (2-3 Marks)

Q1. Explain the difference between Dimensions and Measures in Tableau.

Dimensions are qualitative, categorical fields (like names, regions, or categories) used to group and slice data. Measures are quantitative, numeric values (like profits, sales, or quantities) that can be added, averaged, or mathematically calculated.

Q2. What is the purpose of Data Preprocessing?

Data Preprocessing cleans and structures raw, noisy data before it is fed into an AI model. It involves handling missing values, deleting duplicates, selecting key features, and formatting data, which improves model accuracy and performance.

Math for AI (Statistics & Probability)

Mathematics is the language of machine learning. This unit covers the importance of math in AI pattern matching, explores statistics (mean, median, mode), and details probability principles.

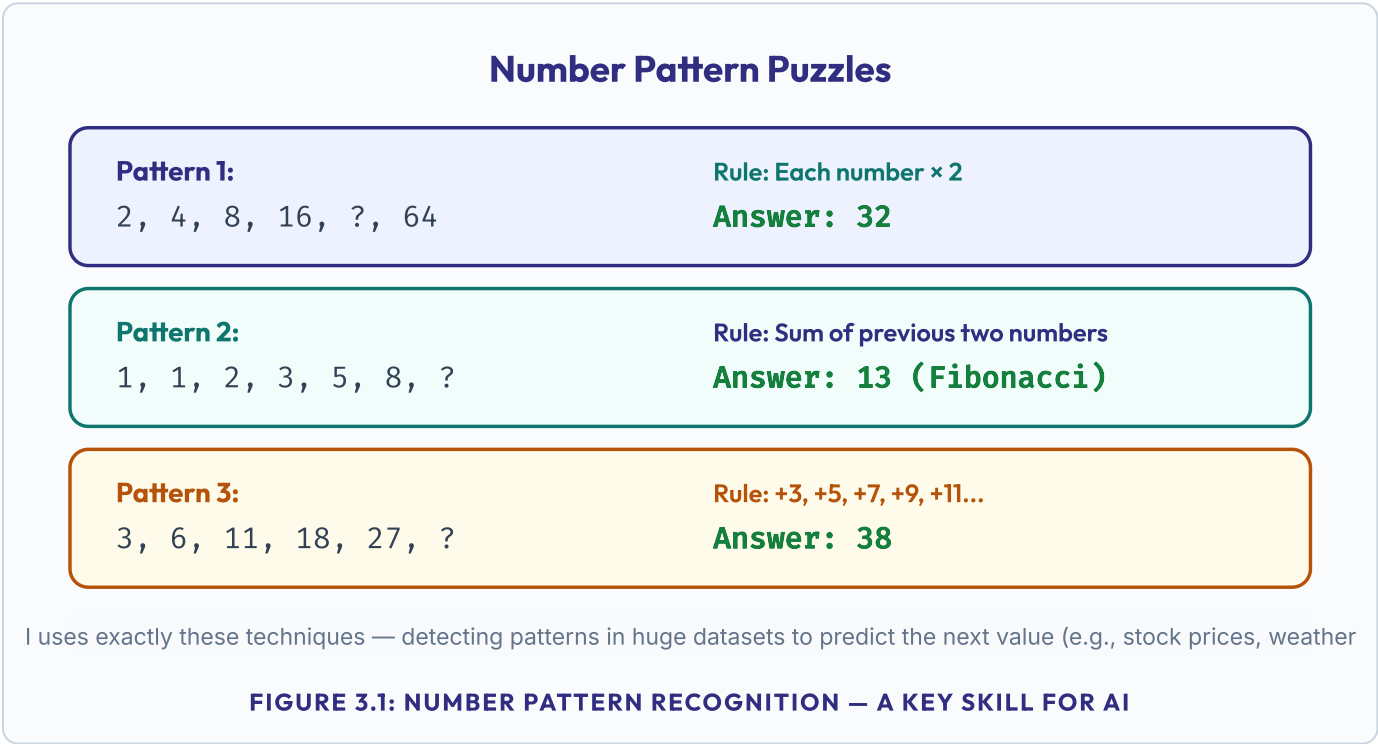
3.1 Importance of Math for AI

Underneath all AI models (like neural networks or regression lines) lie mathematical formulas. Computers do not understand images, speech, or text directly; they convert them into matrices of numbers.

- **Linear Algebra:** Used to represent data, images, and weights in matrices.
- **Calculus:** Helps in optimization (minimizing prediction error during model training).
- **Statistics & Probability:** Used to deal with data trends, uncertainty, and likelihoods.

Activity: Number Patterns

AI systems are, at their core, pattern-recognition engines. The ability to spot patterns in sequences of numbers is a foundational skill. Observe the following sequences and find the missing number:



Activity: Picture Analogy

Picture analogy involves finding connections between sets of images and using that relationship to solve a problem. This skill mirrors how Computer Vision AI finds patterns across image features.

Picture Analogy — Find the Relationship

Analogy 1: Shape Transformation



Analogy 2: Size & Rotation



Computer Vision AI uses similar relational reasoning to classify and transform images.

FIGURE 3.2: PICTURE ANALOGY — IDENTIFYING VISUAL RELATIONSHIPS

TRY IT YOURSELF!

Number Pattern: Find the next number: 5, 10, 20, 40, ?

Picture Analogy: If a large red circle becomes a small red circle, what does a large blue square become?

Think about these — the answers use the same logic that AI pattern-recognition algorithms use!

3.2 Statistics in AI

DEFINITION OF STATISTICS

Statistics is the branch of mathematics that deals with the **collection, organization, analysis, interpretation, and presentation** of numerical data. It helps us make sense of large amounts of information, find patterns, and draw conclusions to support decision-making.

Statistics plays a critical role in AI — every machine learning model relies on statistical methods to learn from data and make predictions.

THREE MEASURES OF CENTRAL TENDENCY

Consider a dataset: [3, 5, 5, 6, 8, 9, 13]

- **Mean (Average):** Sum of all values divided by count.
 - $\text{Mean} = (3+5+5+6+8+9+13) / 7 = 49 / 7 = 7$
- **Median (Middle Value):** The middle value when sorted.
 - In [3, 5, 5, 6, 8, 9, 13] (already sorted, count=7), the 4th element is the median: 6.
- **Mode (Most Frequent):** The value that appears most often.
 - In this dataset, 5 appears twice. Thus, $\text{Mode} = 5$.

Applications of Statistics in Real Life

Real-Life Applications of Statistics



FIGURE 3.4: FOUR KEY APPLICATIONS OF STATISTICS IN REAL LIFE

- **Disaster Management:** Analyzing historic seismic data to predict earthquakes and flood frequencies.
- **Sports:** Analyzing player performance statistics to scout talent and predict match outcomes.
- **Diseases Prediction:** Tracking infection rates, symptoms, and spread patterns to forecast disease outbreaks (e.g., COVID-19 case trend analysis helped governments plan hospital capacity).
- **Weather Forecasting:** Collecting temperature and pressure trends to predict rainfall, storms, and heatwaves.

CAR SPOTTING AND TABULATING ACTIVITY

Imagine standing by a road for 10 minutes and recording the colors of passing cars:

[Red, White, White, Black, White, Red, Black, White, White, Blue]

Tabulation: White: 5, Red: 2, Black: 2, Blue: 1.

Analysis: White is the mode. An AI looking at this data would predict that the next car to pass is most likely to be white.

Car Color Frequency — Bar Chart



FIGURE 3.3: BAR CHART VISUALIZATION OF CAR SPOTTING DATA — WHITE IS THE MODE

ACTIVITY: USES OF STATISTICS IN DAILY LIFE

Students will explore the applications of statistics in real life by collecting and analyzing their own data.

1. **Choose a topic:** Pick something measurable from daily life — e.g., hours of sleep per night, daily water intake (glasses), time spent on homework, or steps walked per day.
2. **Collect data:** Record data for yourself and at least 5 classmates for 5 consecutive days.
3. **Tabulate:** Organize the data in a spreadsheet or table.
4. **Apply statistical measures:**
 - Calculate the **Mean** for each person.
 - Find the **Median** of the class averages.

- Identify the **Mode** — is there a most common value?

5. **Visualize:** Create a bar chart comparing each person's average.

6. **Interpret:** What does the data tell you? Who sleeps the most? Is there a pattern on weekdays vs weekends?

Data collection plays a key role in Artificial Intelligence — it forms the basis of statistics and interpretation by AI. This activity helps you think like a data scientist!

3.3 Probability in AI

Probability measures the likelihood of an event occurring. It is calculated as: $P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$ Probability values range strictly between \$0\$ (impossible event) and \$1\$ (certain event).

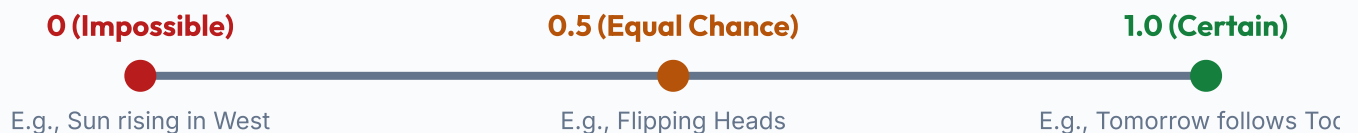
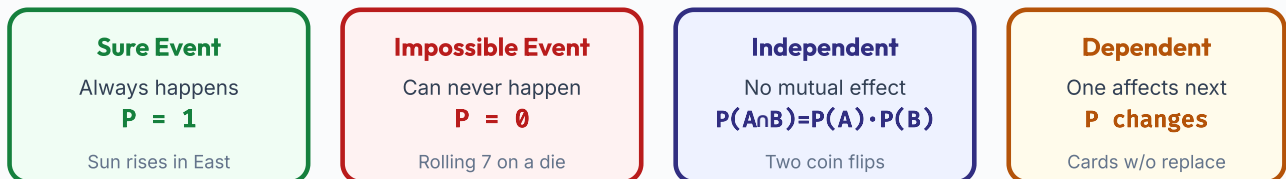


FIGURE 2.6: THE PROBABILITY SPECTRUM AND SAMPLE EVENTS

Types of Probability Events

Types of Probability Events



Real-World Applications of Probability



FIGURE 3.4: TYPES OF PROBABILITY EVENTS AND THEIR REAL-WORLD APPLICATIONS

- Sure Event:** An event that is guaranteed to happen. Probability $P = 1$.
- Impossible Event:** An event that cannot happen. Probability $P = 0$.
- Independent Events:** The outcome of one event does not affect another (e.g., rolling a 6 on a die, then rolling a 6 again).
- Dependent Events:** The outcome of the first event affects the next (e.g., drawing a red card from a deck of cards and not replacing it before drawing a second card).

EXERCISE: IDENTIFY THE TYPE OF EVENT

Classify each of the following as a **Sure Event**, **Impossible Event**, **Independent Event**, or **Dependent Event**:

1. Getting a number less than 7 when rolling a standard die.

Answer: Sure Event ($P=1$, all faces are 1-6)

2. Drawing a green ball from a bag containing only red and blue balls.

Answer: Impossible Event ($P=0$)

3. Flipping a coin and getting heads, then flipping it again and getting tails.

Answer: Independent Events

4. Drawing two cards from a deck without replacement — drawing a King first affects the probability of drawing a King second.

Answer: Dependent Events

5. It will rain tomorrow AND your school cricket team wins a match.

Answer: Independent Events

6. Getting a number 8 on a standard die.

Answer: Impossible Event ($P=0$, die has only 1-6)

EXERCISE: REVISION TIME — PROBABILITY PRACTICE

Solve the following problems to revise all probability concepts:

1. **Coin Toss:** A coin is tossed twice. What is the probability of getting at least one head?

Total outcomes = {HH, HT, TH, TT} = 4. Favorable (at least one head) = {HH, HT, TH} = 3. $P = 3/4 = 0.75$

2. **Die Roll:** A die is rolled once. What is the probability of getting an even number?

Even numbers on a die = {2, 4, 6} = 3. Total outcomes = 6. $P = 3/6 = 1/2 = 0.5$

3. **Card Draw:** A card is drawn from a standard deck of 52 cards. What is the probability that it is a heart?

Hearts in a deck = 13. Total cards = 52. $P = 13/52 = 1/4 = 0.25$

4. **Application — Weather:** A weather app says there is a 70% chance of rain tomorrow. What is the probability it will NOT rain?

$P(\text{no rain}) = 1 - P(\text{rain}) = 1 - 0.70 = 0.30$ or 30%

5. **Application — Traffic:** Based on past data, the probability of heavy traffic on Monday morning is 0.6 and on Saturday morning is 0.2. On which day should you leave earlier for school?

Monday — it has a higher probability (0.6 vs 0.2) of heavy traffic, so you should leave earlier on Mondays.

Unit 3 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. What is the median of the following set of marks: [12, 15, 18, 20, 22]?

- a) 15
- b) 18
- c) 20
- d) 17.4

Answer: b

Q2. A bag has 3 red balls and 7 blue balls. What is the probability of drawing a red ball?

- a) 0.7
- b) 0.3
- c) 0.5
- d) 1.0

Answer: b

Short Answer Questions (2-3 Marks)

Q1. Calculate the mean, median, and mode for this dataset: [4, 8, 6, 8, 14].

First, sort the data: [4, 6, 8, 8, 14].

- **Mean:** $(4+6+8+8+14)/5 = 40/5 = 8$.

- **Median:** The middle (3rd) value is 8.

- **Mode:** The number 8 appears most frequently (twice).

Q2. Explain independent vs dependent events in probability with examples.

Independent events do not influence each other's likelihood (e.g., flipping heads on a coin twice in a row; the first flip does not change the 50% chance of the second). Dependent events are influenced by previous outcomes (e.g., pulling a marble from a jar and keeping it out, which changes the total count and probabilities for subsequent draws).

Introduction to Generative AI

Generative AI marks a leap in computer capability, moving from analyzing existing data to creating brand-new artifacts. This unit contrasts Generative and Conventional AI, explains how GANs function, and examines ethical issues like copyright and deepfakes.

4.1 Concept & Differences from Conventional AI

For decades, computers were designed to analyze, classify, and filter data. They could tell you if an image contained a cat, but they could not draw a new cat from scratch. Generative AI changes this.

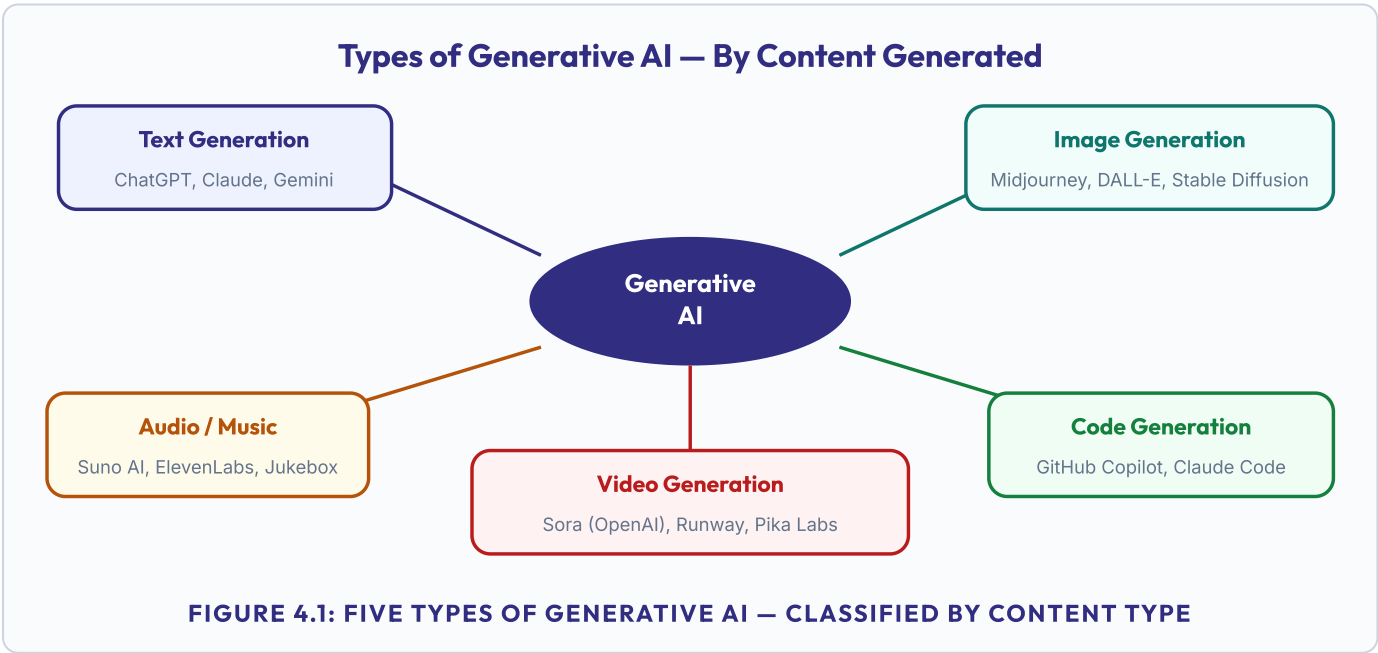
GENERATIVE AI DEFINITION
Generative AI (GenAI) is a category of artificial intelligence models that can generate new content—including text, images, audio, video, and computer code—in response to prompts, by learning patterns from massive training datasets.

Generative AI vs. Conventional AI

FEATURE	CONVENTIONAL AI	GENERATIVE AI
Primary Goal	Analyze, classify, predict, or make decisions based on inputs.	Create new, original content that resembles the training data.
Output Type	Numbers, classifications, probabilities (e.g., "Spam/Not Spam").	Images, text paragraphs, audio songs, code scripts.
Examples	Spam filters, credit card fraud detection, stock price predictors.	ChatGPT (text), Midjourney (images), GitHub Copilot (code).

Types of Generative AI

Generative AI is not limited to one kind of content. It spans multiple media types, each powered by specialized models:



Examples of Generative AI in Action

TYPE	TOOL / MODEL	WHAT IT DOES
Text	ChatGPT, Claude	Generates essays, stories, summaries, answers questions, writes emails, and explains concepts in natural language.
Image	DALL-E, Midjourney	Creates original images from text descriptions (e.g., "a cat wearing a space suit on Mars").
Audio	Suno AI, ElevenLabs	Generates music tracks, voice clones, and realistic speech from text prompts.
Video	Sora, Runway	Creates short video clips from text descriptions or extends existing video footage.
Code	GitHub Copilot	Writes, completes, and debugs programming code based on comments or partial code.

ACTIVITY: GUESS THE REAL IMAGE VS. THE AI-GENERATED IMAGE

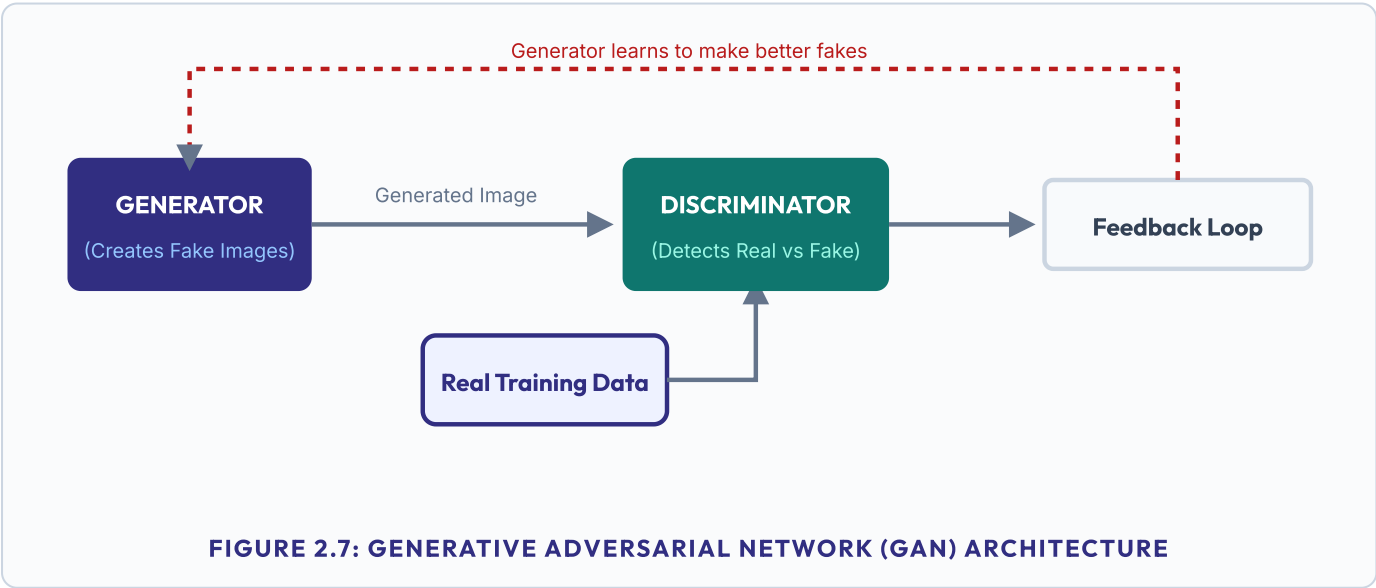
Can you tell which images are real and which are AI-generated? Test your skills:

1. Your teacher will show 10 pairs of images — one real photograph and one AI-generated image (created using tools like Midjourney or DALL-E).
2. For each pair, vote: **Which one is real?** Record your answers on paper.
3. After all 10 pairs, reveal the answers. Count how many you got right.
4. **Class Discussion:**
 - What visual clues helped you spot the fakes? (Look for: distorted hands, unnatural text, weird backgrounds, asymmetric earrings, melting edges.)
 - Were some AI images so realistic that they fooled the entire class?
 - What does this mean for trusting images online?

This activity demonstrates why media literacy and critical thinking are essential in the age of Generative AI.

4.2 How Generative AI Works: GANs

One of the primary frameworks driving image generation is the **Generative Adversarial Network (GAN)**.



- A GAN consists of two neural networks competing in a game:
- **The Generator:** Attempts to create realistic synthetic data (e.g., fake images of human faces) starting from random noise.
 - **The Discriminator:** Evaluates the generated image against real training images and guesses whether it is "Real" or "Fake".

- **The Game:** The generator learns to make better fakes to trick the discriminator, and the discriminator learns to detect fakes better. Over time, the generator produces highly realistic outputs.

HANDS-ON ACTIVITY: GAN PAINT STUDIO

Experience how GANs work by trying **GAN Paint Studio** — a tool created by MIT that lets you paint objects into scenes using a GAN.

1. Open GAN Paint Studio (search "GAN Paint Studio MIT" in your browser).
2. Select a scene (e.g., a church, a kitchen, or a landscape).
3. Use the brush tools to **add** objects like trees, grass, doors, or clouds into the image.
4. Use the **remove** tool to erase objects from the scene.
5. Observe how the GAN automatically blends your additions into the scene realistically!

What to notice: The GAN understands the *structure* of the scene — it won't place a tree in the sky or grass on a ceiling. This shows the model has learned spatial relationships from its training data.

Generative AI Tools for Students to Explore

Try these free or beginner-friendly Generative AI tools to create your own content:

Generative AI Tools You Can Try

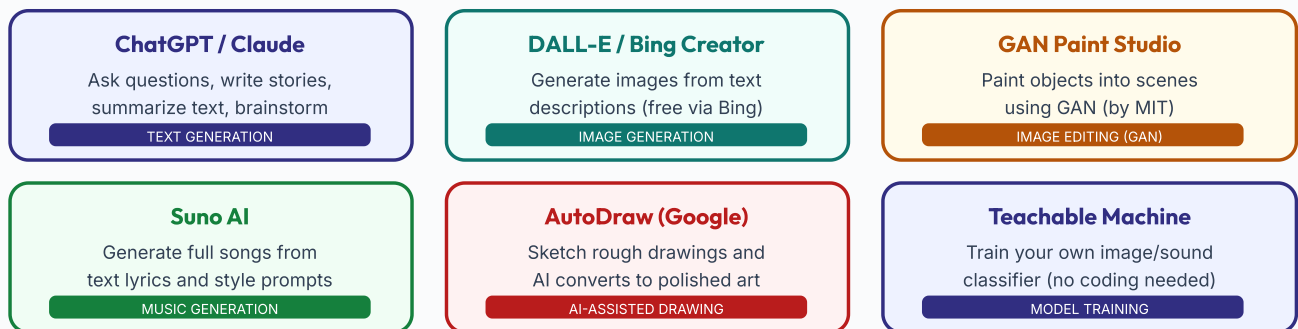


FIGURE 4.3: GENERATIVE AI TOOLS STUDENTS CAN EXPLORE

CLASSROOM ACTIVITY: EXPLORE GENAI TOOLS

In groups of 3-4, try one of the tools above and present your experience to the class:

- **Group 1:** Use ChatGPT or Claude to write a short poem about AI, then ask it to rewrite the poem in a different style.
- **Group 2:** Use Bing Image Creator (DALL-E) to generate an image of "a futuristic classroom in India." Discuss: Does the output match your imagination?
- **Group 3:** Use GAN Paint Studio to add and remove objects from a scene. Report: What surprised you about how the AI blends objects?
- **Group 4:** Use AutoDraw to sketch 5 objects. How accurately did the AI guess what you were drawing?

4.3 Benefits, Limitations, and Ethical Concerns

Benefits

- **Creativity Support:** Helps writers, designers, and programmers brainstorm ideas.
- **Automation:** Drafts emails, customer service responses, and basic code.
- **Synthetic Data:** Generates medical scans to train diagnostics AI safely.

Limitations

- **Hallucinations:** GenAI can state false or inaccurate information confidently.
- **Lack of Real Understanding:** GenAI models match statistical patterns; they do not comprehend meaning.

- **High Computational Cost:** Training large models requires enormous energy and specialized hardware (GPUs/TPUs).
- **Data Dependency:** Quality of output depends entirely on the quality and diversity of training data.
- **No Fact-Checking:** GenAI cannot verify if the content it generates is factually correct.

Benefits vs. Limitations of Generative AI

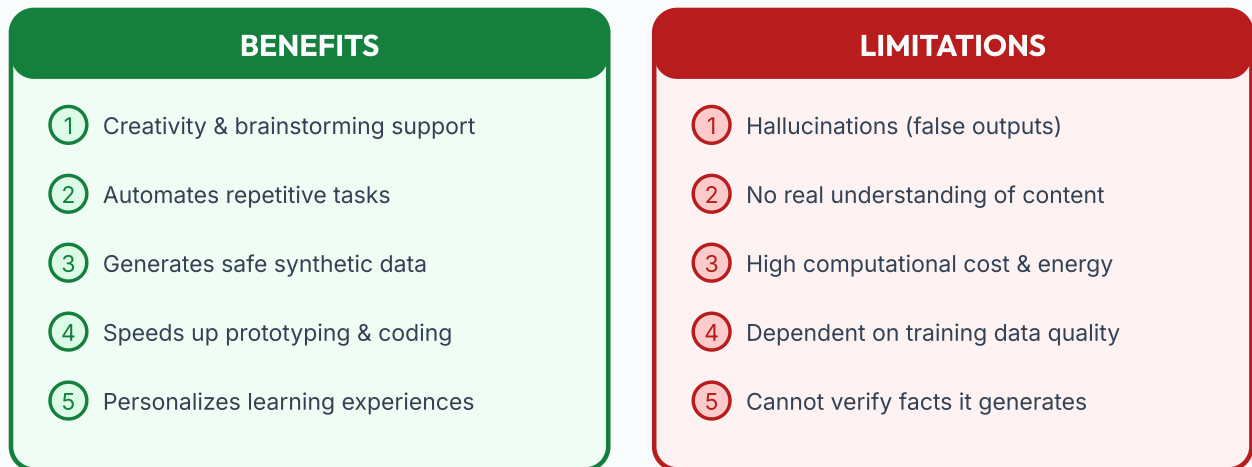


FIGURE 4.2: BENEFITS VS. LIMITATIONS OF GENERATIVE AI – A BALANCED VIEW

ETHICAL RISKS OF GENAI

- **Deepfakes:** Realistic video/image manipulations depicting people saying or doing things they never did.
- **Copyright Infringement:** GenAI models are trained on artists' work without permission or compensation.
- **Spreading Misinformation:** The ease of generating believable text makes it easy to flood the internet with fake news.

Unit 4 Review Questions & Exercises

Multiple Choice Questions (MCQs)

- Q1.** Which network in a GAN architecture learns to generate fake data from random noise?
- Discriminator
 - Generator
 - Predictor
 - Regulator

Answer: b

- Q2.** The tendency of Generative AI to output false information confidently is called:
- Bias replication
 - Preprocessing error
 - Hallucination
 - Network feedback

Answer: c

Short Answer Questions (2-3 Marks)

- Q1.** Explain the role of the Discriminator in a GAN.

The Discriminator acts as an evaluator or judge. It is fed both real images from the dataset and fake images created by the Generator. Its task is to classify whether each input is real or fake, providing feedback that helps the Generator improve.

- Q2.** List two ethical concerns related to Generative AI.

(1) The creation of deepfakes, which manipulate audio and video to impersonate individuals without consent, and (2) Copyright violations, where models are trained on intellectual property without licensing.

Introduction to Python (Theory)

Python is the standard language of AI and data science due to its clean syntax and readability. This unit introduces variables, data types, operators, conditionals, iteration, and list operations in Python.

WARM-UP ACTIVITY: LEARN PROGRAMMING THROUGH GAMES!

Before diving into Python syntax, try **Code Combat** (<https://codecombat.com/>) — a gamified platform where you write real code to control a hero through dungeons. It teaches programming logic (variables, loops, conditionals) through fun game levels. This builds your coding intuition before writing Python scripts!

5.1 Introduction & Core Syntax

Created by **Guido van Rossum** in 1991, Python is an interpreted, high-level, and general-purpose programming language.

WHY IS PYTHON USED IN AI?

- **Easy to Learn:** Readability makes it accessible for beginners.
- **Rich Libraries:** Built-in packages for statistics, math, and AI (e.g., NumPy, Pandas, TensorFlow).
- **Platform Independent:** Runs on Windows, Mac, and Linux.

Applications of Python

Where is Python Used?

AI & Machine Learning

ChatGPT, self-driving cars, face recognition

Data Science & Analytics

Analyzing trends, dashboards, reports

Web Development

Instagram, Spotify, YouTube (backend)

Game Development

Pygame library, simulations, quizzes

Automation & Scripting

Auto-send emails, rename files, bots

FIGURE B5.3: TOP 5 APPLICATION AREAS OF PYTHON PROGRAMMING

Variables and Data Types

A **variable** is a named memory location used to store data values. In Python, variables are created the moment you assign a value to them:

PYTHON VARIABLE DECLARATION

```
x = 10          # Integer (int)
y = 5.7         # Floating Point (float)
name = "AI"     # String (str)
is_ready = True # Boolean (bool)
```

Python Data Types at a Glance



Type Conversion: `int("5") → 5` | `float("3.14") → 3.14` | `str(10) → "10"`
Use `int()`, `float()`, `str()` functions to convert between data types.

FIGURE B5.1: PYTHON DATA TYPES AND TYPE CONVERSION

Python Operators

- **Arithmetic Operators:** + (Addition), - (Subtraction), * (Multiplication), / (Division), % (Modulus/Remainder), ** (Exponentiation/Power).
- **Comparison Operators:** == (Equal to), != (Not equal to), > (Greater than), < (Less than), >=, <=.
- **Logical Operators:** and, or, not. These combine multiple conditions:
and — Both conditions must be True. or — At least one must be True. not — Reverses the result.
Example: `if age >= 18 and has_id == True:`
- **Assignment Operators:** = (Assign), += (Add & assign), -= (Subtract & assign), *=, /=.

Expressions

An **expression** is a combination of values, variables, and operators that Python evaluates to produce a result. Every time Python calculates something, it is evaluating an expression.

PYTHON EXPRESSIONS

```
# Arithmetic expression
result = 5 + 3 * 2    # result = 11 (multiplication first, then addition)

# String expression
greeting = "Hello" + " " + "World" # greeting = "Hello World" (concatenation)

# Comparison expression (returns True/False)
is_adult = age >= 18  # True if age is 18 or more

# Logical expression
can_drive = age >= 18 and has_license == True
```

The print() Function

The `print()` function displays output on the screen. It is the most commonly used function in Python.

USING PRINT()

```
# Print a string
print("Hello, World!")    # Output: Hello, World!

# Print a variable
name = "Aarav"
print("My name is", name) # Output: My name is Aarav

# Print with calculation
print("Sum =", 10 + 20)    # Output: Sum = 30
```

The input() Function

The `input()` function takes input from the user via the keyboard. It **always returns a string**, so you must convert it to the required data type using type conversion.

USING INPUT() WITH TYPE CONVERSION

```
# Taking string input
name = input("Enter your name: ")
print("Hello,", name)

# Taking integer input (must convert from string)
age = int(input("Enter your age: "))
print("You are", age, "years old")

# Taking float input
marks = float(input("Enter marks: "))
print("Your marks:", marks)
```

COMMON MISTAKE: FORGETTING TYPE CONVERSION

`input()` always returns a **string**. If you write `num = input("Enter number: ")` and the user types 5, then `num` is the string "5", not the integer 5. You cannot do math with it until you convert: `num = int(input("Enter number: "))`.

5.2 Flow of Control & Loops

By default, a computer executes code line by line. We can control this flow using conditions and loops.

Python Control Flow — Three Execution Patterns

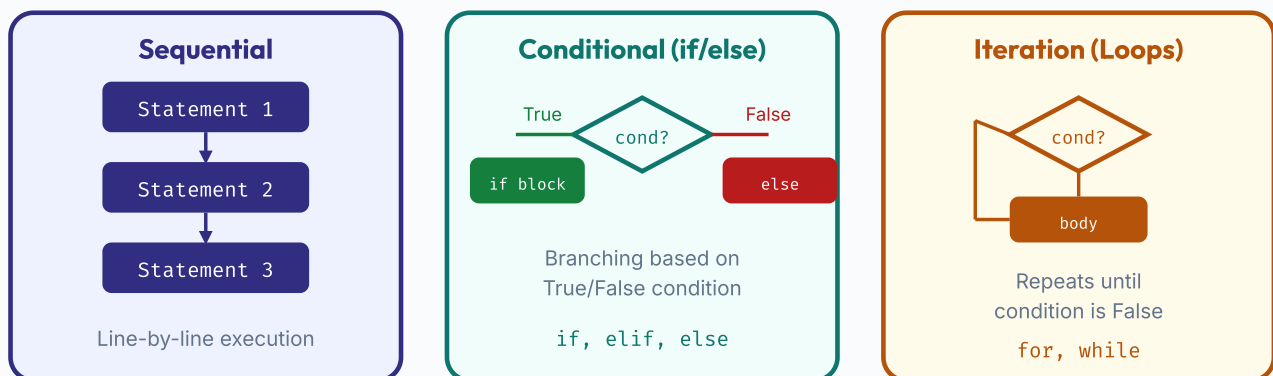


FIGURE B5.2: THREE TYPES OF CONTROL FLOW IN PYTHON

Conditional Statements

Python supports `if`, `elif` (else if), and `else` to create branching logic:

IF / ELSE — SIMPLE CONDITION

```
if age >= 18:
    print("You can vote!")
else:
    print("You are too young.")
```

IF / ELIF / ELSE — MULTIPLE CONDITIONS (GRADE CHECKER)

```
marks = int(input("Enter marks: "))

if marks >= 90:
    print("Grade: A+")
elif marks >= 75:
    print("Grade: A")
elif marks >= 60:
    print("Grade: B")
elif marks >= 40:
    print("Grade: C")
else:
    print("Grade: F (Fail)")
```

HOW ELIF WORKS

Python checks each condition **top to bottom**. The moment one condition is True, it runs that block and skips the rest. If no condition is True, the else block runs. You can have as many elif blocks as needed.

Looping Statements

- **for loop:** Used to iterate over a sequence (like a list or range of numbers).

```
for i in range(1, 4):
    print(i) # Prints 1, 2, 3
```

- **while loop:** Repeats a block of code as long as a condition is true.

```
count = 1
while count <= 3:
    print(count)
    count = count + 1
```

5.3 Python Lists

A **list** is an ordered, mutable sequence of items. It can store multiple data types under a single variable name.

LIST DECLARATION AND OPERATIONS

```
fruits = ["apple", "banana", "cherry"]

# Accessing elements (Indexing starts at 0)
print(fruits[0]) # Output: apple

# Adding element to end
fruits.append("orange")

# Removing element
fruits.remove("banana")

# Getting length of list
print(len(fruits)) # Output: 3
```

List Slicing

Slicing lets you extract a portion of a list using the syntax `list[start:end]`. The start index is included, but the end index is excluded.

LIST SLICING EXAMPLES

```
numbers = [10, 20, 30, 40, 50, 60]

# Positive indexing: elements from index 1 to 3 (4 excluded)
print(numbers[1:4])    # Output: [20, 30, 40]

# Negative indexing: last 3 elements
print(numbers[-3:])    # Output: [40, 50, 60]

# From start to index 3
print(numbers[:3])     # Output: [10, 20, 30]

# Sorting a list
numbers.sort()
print(numbers)         # Output: [10, 20, 30, 40, 50, 60]

# Extending a list with another list
numbers.extend([70, 80])
print(numbers)         # Output: [10, 20, 30, 40, 50, 60, 70, 80]
```

PRACTICE EXERCISES – TRY ON PYTHON COMPILER

Open a Python compiler (IDLE, VS Code, or an online compiler like replit.com) and try these:

1. Create a variable name using `input()`, then print "Hello, [name]! Welcome to AI class."
2. Take two numbers as input, calculate their sum, difference, product, and quotient. Print all four results.
3. Write an `if/elif/else` program that checks if a number entered by the user is positive, negative, or zero.
4. Use a `for` loop to print the first 10 even numbers (2, 4, 6, ... 20).
5. Create a list of 5 fruits. Use `append()` to add one more, `remove()` to delete one, and print the final list with its length.
6. Create a list [5, 15, 25, 35, 45] and print elements from index 1 to 3 using slicing.

UNIT SUMMARY

Python basics are fundamental for Part C (Practical Work). Remember that indexing starts at 0, indentation is syntactically mandatory to define blocks of code, `input()` always returns a string (convert with `int()`/`float()`), and variables do not require explicit type declaration before assignment.

Unit 5 Review Questions & Exercises

Multiple Choice Questions (MCQs)

Q1. Which operator is used to calculate the remainder of a division in Python?

- a) /
- b) %
- c) //
- d) **

Answer: b

Q2. If `numbers = [10, 20, 30]`, what will `numbers[1]` return?

- a) 10
- b) 20
- c) 30
- d) Error

Answer: b

Short Answer Questions (2-3 Marks)

Q1. Explain the role of indentation in Python.

Unlike other languages (like Java or C++) that use curly braces `{ }` to define code blocks, Python uses whitespace indentation. All statements inside a conditional block or loop must have the same number of spaces/tabs to be executed as part of that block.

Q2. What is the difference between `/` and `%` operators in Python?

The division operator `/` divides the numerator by the denominator and returns the quotient as a float value (e.g., `5 / 2` returns `2.5`). The modulus operator `%` divides the numbers and returns only the integer remainder (e.g., `5 % 2` returns `1`).

The Class 9 AI Practical Examination

The practical exam accounts for 50% of your subject grade (50 out of 100 marks). Out of this, 35 marks are evaluated in the Computer Lab under Part C, and 15 marks are evaluated via the Project Work / Student Portfolio under Part D.

C1.1 Marks Distribution Scheme

The 35 marks for Part C are evaluated during a 2-hour lab examination session as follows:

EVALUATION ITEM	DESCRIPTION	MARKS ALLOCATED
Practical Lab Exam	Writing and executing three Python programs from different sections (Print/Input, List, Conditions/Loops) on the computer compiler.	15 Marks
Practical File (Record)	A physical file maintained by the student throughout the year, containing a minimum of 15 solved Python programs printed out with outputs.	15 Marks
Viva Voce	An oral examination conducted by the internal/external examiner asking fundamental Python, AI, and project-based questions.	05 Marks
Total Part C Marks	Evaluated in the computer laboratory.	35 Marks

What to Expect on Exam Day

- Seat Assignment:** Students are assigned a terminal in the computer lab. Use Python IDLE, Anaconda Jupyter Notebook, or a specified IDE.
- Question Sheet:** You will receive a sheet with 3 programming tasks (e.g., Program 1: Input & Arithmetic; Program 2: List Operations; Program 3: If-Else or Loop logic).
- Implementation:** You must write the code on the computer, save it, compile it, and show the working output to the examiner.
- Answer Sheet Writing:** You must write down the algorithm/code and expected output on your physical answer booklet.
- Viva Session:** The examiner will call you one-by-one to inspect your practical file and ask oral questions.

Python Basics & Control Flow Concept Guide

Before coding, we must understand the core constructs of Python. This guide details variable scopes, list indexing rules, conditional flow branches, and loop execution.

C2.1 List Indexing Deep Dive

Lists in Python are ordered sequences. Every item in a list has a position index. Python supports two types of indexing:

- **Positive Indexing:** Starts from 0 at the left and increases to the right.
- **Negative Indexing:** Starts from -1 at the right and decreases to the left.

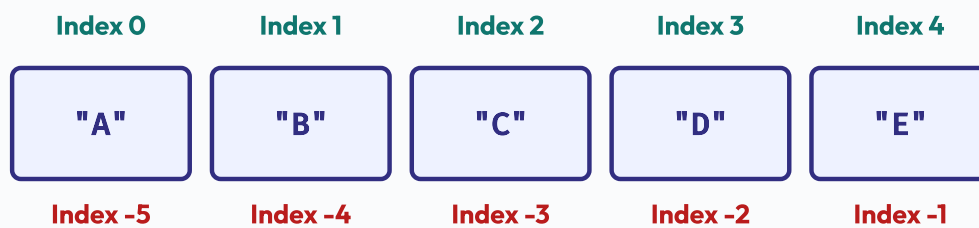


FIGURE 3.1: POSITIVE VS. NEGATIVE LIST INDEXING IN PYTHON

SLICING SYNTAX

Slicing syntax is: `list_name[start : stop : step]`. Note that the stop index is **exclusive**—it slices up to but does not include that index.

Example: `items[1:4]` extracts items at index 1, 2, and 3.

C2.2 Basic Control Flow branching

Programs perform decision-making through conditional branches:

- **if Statement:** Runs a block only if the test condition is True.
- **elif Statement:** Stands for "else if". Checks another condition if the previous one is False.
- **else Statement:** Runs if all previous conditions evaluate to False.

C2.3 Iteration (Loops) logic

- **for loop:** Best when you know the number of iterations in advance (e.g., printing numbers 1 to 10 or looping over a predefined list).
- **while loop:** Best when repeating an action until a condition changes (e.g., asking for a password until the input matches). Requires updating a counter inside the loop to avoid an **infinite loop**.

24 Solved Python Programs

This section contains the official solved programs for Class 9 AI lab files. Each program features a detailed analysis breakdown to help you master Python logic.

Program 1: Personal Information Print

Aim: To print personal information like Name, Father's Name, Class, and School Name on the console.

CONCEPT

The `print()` function displays text on the screen. Commas or formatting can separate text items. Each call to `print()` automatically adds a newline character at the end.

PYTHON SOURCE CODE

```
# Program to print personal details
print("Name: Rohan Sharma")
print("Father's Name: Mr. Suresh Sharma")
print("Class: IX-A")
print("School Name: Green Meadows Public School")
```

EXPECTED CONSOLE OUTPUT

Name: Rohan Sharma Father's Name: Mr. Suresh Sharma Class: IX-A School Name: Green Meadows Public School

Line-by-Line Explanation:

- Line 1: A comment line starting with #, ignored by Python.
- Lines 2-5: Multiple `print()` statements, each displaying the specified text string on a new line.

COMMON MISTAKE & EXAM TIP

Mistake: Typing `Print()` with a capital 'P'. Python is case-sensitive; keywords must be exact.

Exam Tip: You can print empty lines using an empty `print()` call.

Program 2: Print Patterns

Aim: To print a right-angled triangle pattern of asterisks using multiple print statements.

CONCEPT

Using structured strings within `print()` commands to align characters on the screen, creating geometric designs.

PYTHON SOURCE CODE — PATTERN 1: ASCENDING TRIANGLE

```
# Pattern 1: Ascending right-angled triangle
print("*")
print("* ")
print("* * ")
print("* * * ")
print("* * * * ")
print("* * * * * ")
```

EXPECTED CONSOLE OUTPUT — PATTERN 1

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

PYTHON SOURCE CODE — PATTERN 2: DESCENDING TRIANGLE

```
# Pattern 2: Descending (inverted) right-angled triangle
print("* * * * *")
print("* * * *")
print("* * *")
print("* *")
print("*")
```

EXPECTED CONSOLE OUTPUT — PATTERN 2

```
* * * * *
* * * *
* * *
* *
*
```

Explanation: Pattern 1 — Each subsequent print command outputs one more asterisk than the previous, resulting in an ascending right-angled triangle of height 5.

Pattern 2 — The reverse: each line has one fewer asterisk, resulting in a descending (inverted) triangle. Both patterns are created using simple print() statements with no loops required.

DEBUGGING TIP

Make sure each pattern line is enclosed in quotes (single or double). Leaving quotes out causes a syntax error.

Program 3: Square of Number 7

Aim: To calculate and print the square of the number 7 using variables and arithmetic operators.

CONCEPT

We can calculate squares using multiplication (*) or the exponentiation operator (**).

PYTHON SOURCE CODE

```
# Calculate square of 7
num = 7
square = num ** 2
print("The square of 7 is:", square)
```

EXPECTED CONSOLE OUTPUT

```
The square of 7 is: 49
```

Explanation: We assign 7 to num, raise it to the power of 2 using num ** 2, store the result in square, and print it.

Program 4: Sum of Two Numbers

Aim: To calculate the sum of two pre-defined numbers (15 and 20) and print the result.

CONCEPT

Addition is performed using the + operator. We assign values to variables, add them, and print the sum.

PYTHON SOURCE CODE

```
num1 = 15
num2 = 20
total = num1 + num2
print("The sum of 15 and 20 is:", total)
```

EXPECTED CONSOLE OUTPUT

```
The sum of 15 and 20 is: 35
```

Explanation: Stores values in num1 and num2, adds them to calculate the total, and displays the output.

Program 5: Kilometers to Meters Conversion

Aim: To convert a distance given in kilometers into meters.

CONCEPT

Mathematical conversion factor: $1 \text{ km} = 1000 \text{ meters}$.

PYTHON SOURCE CODE

```
km = 4.5
meters = km * 1000
print(km, "kilometers is equal to", meters, "meters")
```

EXPECTED CONSOLE OUTPUT

```
4.5 kilometers is equal to 4500.0 meters
```

Explanation: Multiplies the float variable km by 1000 to convert to meters.

Program 6: Multiplication Table of 5

Aim: To print the multiplication table of 5 up to five terms using simple print statement multiplications.

CONCEPT

Performing arithmetic operations inside print calls to output a table format.

PYTHON SOURCE CODE

```
num = 5
print(num, "x 1 =", num * 1)
print(num, "x 2 =", num * 2)
print(num, "x 3 =", num * 3)
print(num, "x 4 =", num * 4)
print(num, "x 5 =", num * 5)
```

EXPECTED CONSOLE OUTPUT

```
5 x 1 = 5 5 x 2 = 10 5 x 3 = 15 5 x 4 = 20 5 x 5 = 25
```

Program 7: Simple Interest Calculation

Aim: To calculate Simple Interest with Principal = 2000, Rate = 4.5%, and Time = 10 years.

CONCEPT

Simple Interest formula: $SI = (P \times R \times T) / 100$.

PYTHON SOURCE CODE

```
principal = 2000
rate = 4.5
time = 10
si = (principal * rate * time) / 100
print("Principal Amount: Rs.", principal)
print("Rate of Interest:", rate, "%")
print("Time Period:", time, "years")
print("Simple Interest Calculated: Rs.", si)
```

EXPECTED CONSOLE OUTPUT

```
Principal Amount: Rs. 2000 Rate of Interest: 4.5 % Time Period: 10 years Simple Interest Calculated: Rs. 900.0
```

Program 8: Area and Perimeter of Rectangle (Input)

Aim: Input the length and breadth of a rectangle, calculate its Area and Perimeter, and print the results.

CONCEPT

The `input()` function reads inputs as strings. We must convert them to numbers using `float()` or `int()` before performing math operations.

Formulas: $\text{Area} = L \times B$; $\text{Perimeter} = 2 \times (L + B)$.

PYTHON SOURCE CODE

```
# Area and Perimeter of Rectangle with User Input
length = float(input("Enter the length of the rectangle: "))
breadth = float(input("Enter the breadth of the rectangle: "))

area = length * breadth
perimeter = 2 * (length + breadth)

print("Area of the rectangle:", area)
print("Perimeter of the rectangle:", perimeter)
```

EXPECTED CONSOLE OUTPUT

```
Enter the length of the rectangle: 10 Enter the breadth of the rectangle: 5 Area of the rectangle: 50.0 Perimeter
of the rectangle: 30.0
```

Line-by-Line Explanation:

- Line 2: `input()` prompts the user. The input string is passed to `float()` for conversion, then stored in `length`.
- Line 5: Calculates the area by multiplying `length` and `breadth`.
- Line 6: Calculates the perimeter using `2 * (length + breadth)`. Note the parentheses around addition.

COMMON MISTAKE

Forgetting to typecast: `length = input(...)`. If you type `length = "10"` and `breadth = "5"`, calculating `length * breadth` will trigger a `TypeError` because Python cannot multiply strings.

Program 9: Area of Triangle (Input)

Aim: Input base and height of a triangle and calculate its Area.

CONCEPT

Formula: $\text{Area} = 0.5 \times \text{Base} \times \text{Height}$.

PYTHON SOURCE CODE

```
base = float(input("Enter the base of the triangle: "))
height = float(input("Enter the height of the triangle: "))
area = 0.5 * base * height
print("The area of the triangle is:", area)
```

EXPECTED CONSOLE OUTPUT

```
Enter the base of the triangle: 8 Enter the height of the triangle: 12 The area of the triangle is: 48.0
```

Program 10: Average Marks of 3 Subjects (Input)

Aim: Input marks for 3 subjects and calculate their average.

CONCEPT

Average is calculated by adding the numbers together and dividing by the total count.

Formula: $\text{Average} = (S_1 + S_2 + S_3) / 3$.

PYTHON SOURCE CODE

```
sub1 = float(input("Enter marks for Subject 1: "))
sub2 = float(input("Enter marks for Subject 2: "))
sub3 = float(input("Enter marks for Subject 3: "))

average = (sub1 + sub2 + sub3) / 3
print("The average marks are:", average)
```

EXPECTED CONSOLE OUTPUT

```
Enter marks for Subject 1: 85 Enter marks for Subject 2: 90 Enter marks for Subject 3: 95 The average marks are:
90.0
```

EXAM TIP: OPERATOR PRECEDENCE

Always group the addition in parentheses: $(sub1 + sub2 + sub3) / 3$. Writing $sub1 + sub2 + sub3 / 3$ will divide only $sub3$ by 3 first due to operator precedence, producing incorrect results.

Program 11: Discount Calculator

Aim: Input the bill amount and a discount percentage, calculate the discount amount, and print the final payable amount.

CONCEPT

- $\text{Discount Value} = (\text{Bill Amount} \times \text{Discount Percentage}) / 100$.
- $\text{Final Amount} = \text{Bill Amount} - \text{Discount Value}$.

PYTHON SOURCE CODE

```
bill = float(input("Enter total bill amount (Rs): "))
disc_pct = float(input("Enter discount percentage: "))

disc_value = (bill * disc_pct) / 100
final_bill = bill - disc_value

print("Discount Amount: Rs.", disc_value)
print("Final Amount to Pay: Rs.", final_bill)
```

EXPECTED CONSOLE OUTPUT

```
Enter total bill amount (Rs): 1500 Enter discount percentage: 10 Discount Amount: Rs. 150.0 Final Amount to Pay:
Rs. 1350.0
```

Program 12: Surface Area & Volume of Cuboid

Aim: Input length, width, and height of a cuboid, and calculate its Surface Area and Volume.

CONCEPT

Formulas:

- $\text{Volume} = L \times W \times H$.
- $\text{Surface Area} = 2 \times (L \times W + W \times H + H \times L)$.

PYTHON SOURCE CODE

```
l = float(input("Enter length of the cuboid: "))
w = float(input("Enter width of the cuboid: "))
h = float(input("Enter height of the cuboid: "))

volume = l * w * h
surface_area = 2 * (l*w + w*h + h*l)

print("Volume of cuboid:", volume)
print("Total Surface Area:", surface_area)
```

EXPECTED CONSOLE OUTPUT

```
Enter length of the cuboid: 5 Enter width of the cuboid: 3 Enter height of the cuboid: 4 Volume of cuboid: 60.0
Total Surface Area: 94.0
```

Program 13: List Operations on Student Names

Aim: Create a list of science quiz students containing names "Arjun", "Sonakshi", "Vikram", "Sandhya", "Sonal", "Isha", "Kartik" and perform specific mutations in sequence.

CONCEPT

This program tests list mutation methods: `remove()`, `append()`, and `pop()`.

PYTHON SOURCE CODE

```
# Initial list
students = ["Arjun", "Sonakshi", "Vikram", "Sandhya", "Sonal", "Isha", "Kartik"]
print("Initial list:", students)

# Delete "Vikram" from list
students.remove("Vikram")
print("After removing Vikram:", students)

# Add "Jay" at the end
students.append("Jay")
print("After appending Jay:", students)

# Remove the item at the second position (index 1)
removed_item = students.pop(1)
print("Removed item:", removed_item)
print("Final list:", students)
```

EXPECTED CONSOLE OUTPUT

```
Initial list: ['Arjun', 'Sonakshi', 'Vikram', 'Sandhya', 'Sonal', 'Isha', 'Kartik'] After removing Vikram:
['Arjun', 'Sonakshi', 'Sandhya', 'Sonal', 'Isha', 'Kartik'] After appending Jay: ['Arjun', 'Sonakshi', 'Sandhya',
'Sonal', 'Isha', 'Kartik', 'Jay'] Removed item: Sonakshi Final list: ['Arjun', 'Sandhya', 'Sonal', 'Isha',
'Kartik', 'Jay']
```

Line-by-Line Explanation:

- Line 2: Instantiates the list of strings.
- Line 5: `students.remove("Vikram")` searches for the element "Vikram" and deletes it.
- Line 9: `students.append("Jay")` inserts "Jay" at the end of the list.
- Line 13: `students.pop(1)` removes and returns the item at index 1 ("Sonakshi"). Note that the "second position" corresponds to index 1, as indexing starts at 0.

COMMON MISTAKE

Confusing "second position" with index 2. The item at index 2 is actually the third item. Always remember that index = position - 1.

Program 14: Positive & Negative Index Slicing

Aim: Create a list num=[23,12,5,9,65,44], print its length, slice elements from the 2nd to 4th position using positive indexing, and slice elements from the 3rd to 5th position using negative indexing.

CONCEPT

- Length: calculated using len().
- Positive Slicing: index 1 to 4 (exclusive of 4) is written as [1:4].
- Negative Slicing: position 3 (index 2 or -4) to position 5 (index 4 or -2) is written as [-4:-1].

PYTHON SOURCE CODE

```
num = [23, 12, 5, 9, 65, 44]
print("Length of list:", len(num))

# Positive Indexing: Second (index 1) to Fourth (index 3) position
# We write slice index as 1:4 (includes 1, 2, 3)
slice_pos = num[1:4]
print("Slice (positive index 1:4):", slice_pos)

# Negative Indexing: Third (index -4) to Fifth (index -2) position
# We write slice index as -4:-1 (includes -4, -3, -2)
slice_neg = num[-4:-1]
print("Slice (negative index -4:-1):", slice_neg)
```

EXPECTED CONSOLE OUTPUT

```
Length of list: 6 Slice (positive index 1:4): [12, 5, 9] Slice (negative index -4:-1): [5, 9, 65]
```

EXAM TIP: NEGATIVE INDICES MAPPING

For list num=[23,12,5,9,65,44]:

- Positive indices: 0: 23, 1: 12, 2: 5, 3: 9, 4: 65, 5: 44
- Negative indices: -6: 23, -5: 12, -4: 5, -3: 9, -2: 65, -1: 44

Program 15: List of First 10 Even Numbers & Mutate

Aim: Create a list of the first 10 even numbers, add 1 to each item using a loop, and print the resulting list.

CONCEPT

Iterating over a list using index access to modify its elements in-place.

PYTHON SOURCE CODE

```
# Create list of first 10 even numbers
evens = [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
print("Original List:", evens)

# Loop through the list using indices to add 1 to each item
for i in range(len(evens)):
    evens[i] = evens[i] + 1

print("Modified List (added 1):", evens)
```

EXPECTED CONSOLE OUTPUT

```
Original List: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20] Modified List (added 1): [3, 5, 7, 9, 11, 13, 15, 17, 19, 21]
```

Explanation: The loop runs with `i` from 0 to 9. In each iteration, `evens[i] = evens[i] + 1` reads the element, increments it, and saves it back to the same index.

Program 16: List Extend and Sort

Aim: Create a list `List_1=[10,20,30,40]`, add elements `[14,15,12]` using the `extend()` function, and print the sorted list in ascending order.

CONCEPT

- `extend()`: Appends multiple elements from another list to the end.
- `sort()`: Sorts list items in-place in ascending order.

PYTHON SOURCE CODE

```
List_1 = [10, 20, 30, 40]
print("Original List:", List_1)

# Extend the list with new elements
List_1.extend([14, 15, 12])
print("Extended List:", List_1)

# Sort the list
List_1.sort()
print("Sorted List:", List_1)
```

EXPECTED CONSOLE OUTPUT

```
Original List: [10, 20, 30, 40] Extended List: [10, 20, 30, 40, 14, 15, 12] Sorted List: [10, 12, 14, 15, 20, 30, 40]
```

Program 17: Voting Eligibility Checker

Aim: Input age and check whether a person is eligible to vote (`age >= 18`).

CONCEPT

Using the `if-else` conditional statement to control the flow of execution based on comparison.

PYTHON SOURCE CODE

```
age = int(input("Enter your age: "))

if age >= 18:
    print("Eligible to vote!")
else:
    years_left = 18 - age
    print("Not eligible. Please wait", years_left, "more years.")
```

EXPECTED CONSOLE OUTPUT

```
Enter your age: 15 Not eligible. Please wait 3 more years.
```

Program 18: Student Grade Calculator

Aim: Input a student's marks and calculate their grade based on specific criteria.

CONCEPT

Multi-condition branching using if-elif-else:

- Marks ≥ 90 : Grade A
- Marks 75 to 89: Grade B
- Marks 50 to 74: Grade C
- Marks < 50 : Grade D (Fail)

PYTHON SOURCE CODE

```
marks = float(input("Enter student marks (0-100): "))

if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 50:
    grade = "C"
else:
    grade = "D"

print("Student Grade:", grade)
```

EXPECTED CONSOLE OUTPUT

```
Enter student marks (0-100): 82.5 Student Grade: B
```

Program 19: Positive, Negative, or Zero Checker

Aim: Input a number and verify whether it is positive, negative, or zero.

CONCEPT

Comparison signs: > 0 is positive, < 0 is negative, and $= 0$ is zero.

PYTHON SOURCE CODE

```
num = float(input("Enter a number: "))

if num > 0:
    print("The number is Positive.")
elif num < 0:
    print("The number is Negative.")
else:
    print("The number is Zero.")
```

EXPECTED CONSOLE OUTPUT

```
Enter a number: -12.4 The number is Negative.
```

Program 20: Print First 10 Natural Numbers

Aim: Print numbers from 1 to 10 on the console using a loop.

CONCEPT

The range(start, stop) function generates numbers starting from start up to but not including stop.

PYTHON SOURCE CODE

```
# Loop to print 1 to 10
for i in range(1, 11):
    print(i)
```

EXPECTED CONSOLE OUTPUT

```
1 2 3 ... 10
```

Explanation: `range(1, 11)` generates values 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. The loop assigns each value to `i` and prints it.

Program 21: Print First 10 Even Numbers

Aim: Print the first 10 even numbers (2 to 20) using loops.

CONCEPT

We can use a step argument in `range(start, stop, step)`.

PYTHON SOURCE CODE

```
# range(start, stop, step)
for i in range(2, 21, 2):
    print(i)
```

EXPECTED CONSOLE OUTPUT

```
2 4 6 8 10 12 14 16 18 20
```

Program 22: Print Odd Numbers up to n

Aim: Input a limit `n` and print all odd numbers from 1 to `n`.

PYTHON SOURCE CODE

```
n = int(input("Enter the limit n: "))

print("Odd numbers from 1 to", n, ":")
for i in range(1, n + 1, 2):
    print(i)
```

EXPECTED CONSOLE OUTPUT

```
Enter the limit n: 7 Odd numbers from 1 to 7 : 1 3 5 7
```

Program 23: Sum of First 10 Natural Numbers

Aim: Calculate and print the sum of numbers from 1 to 10.

CONCEPT

We initialize an accumulator variable `total_sum = 0`, loop through the numbers, add each to the accumulator, and print the total sum.

PYTHON SOURCE CODE

```
total_sum = 0

for i in range(1, 11):
    total_sum = total_sum + i

print("The sum of first 10 natural numbers is:", total_sum)
```

EXPECTED CONSOLE OUTPUT

```
The sum of first 10 natural numbers is: 55
```

Program 24: Sum of All Numbers in a List

Aim: Find the sum of all numerical elements stored in a list without using built-in sum functions.

CONCEPT

Iterating over list elements using a for-in loop, adding each element to an accumulator variable.

PYTHON SOURCE CODE

```
numbers = [5, 10, 15, 20, 25]
sum_val = 0

for num in numbers:
    sum_val = sum_val + num

print("The sum of numbers in the list is:", sum_val)
```

EXPECTED CONSOLE OUTPUT

The sum of numbers in the list is: 75

Explanation: The loop directly iterates over each element in the list numbers. The value of num becomes 5, then 10, then 15, and so on, adding each value to sum_val.

IMPORTANT LINKS FOR PYTHON PRACTICE

- **CBSE Python Content Manual (Official):**
https://cbseacademic.nic.in/web_material/Curriculum21/publication/secondary/Python_Content_Manual.pdf
- **CBSE AI Resources Folder (Google Drive):**
<https://drive.google.com/drive/folders/1qRAckDcuIA5i164OUFDIilxb8mT65MMb>

These resources contain additional exercises, reference material, and step-by-step Python guides provided by CBSE for Class 9 AI students.

Viva Voce Question Bank

This question bank contains the 20 most common oral questions asked during Class 9 AI practical evaluations.

Q1. Who developed the Python programming language and when?

Python was developed by Guido van Rossum and was first released in 1991.

Q2. Why is Python called an "interpreted" language?

Unlike compiled languages where code is translated to machine language all at once, Python code is executed line-by-line by an interpreter. If an error is encountered on line 4, execution stops there.

Q3. What is the difference between mutable and immutable data types in Python?

Mutable data types can be modified after they are created (e.g., Lists). Immutable data types cannot be changed once defined; any modification creates a new object in memory (e.g., Strings, Integers, Tuples).

Q4. What is a variable? Do we need to declare its type in Python?

A variable is a named memory location used to store data. In Python, we do not declare variable types explicitly; Python is dynamically typed and automatically determines the type based on the value assigned.

Q5. What is the difference between = and == in Python?

The single equals sign = is the assignment operator used to store a value in a variable. The double equals sign == is a comparison operator used to check if two values are equal, returning a Boolean value (True or False).

Q6. What will be the output of print("AI" * 3)?

The output will be AIAIAI. The multiplication operator * when used with a string replicates it the specified number of times.

Q7. What is a syntax error? Give an example.

A syntax error occurs when you violate the grammar rules of Python. Example: Writing `if x = 5:` instead of `if x == 5:`, or forgetting the colon at the end of an `if` statement.

Q8. How do you write comments in Python? Why are they used?

Comments are written using the hash character #. They are ignored by the compiler and are used to explain the code, making it easier for humans to read and maintain.

Q9. What is the index of the first and last element in a Python list of length n?

The first element has positive index 0 (or negative index -n). The last element has positive index n - 1 (or negative index -1).

Q10. What is an infinite loop? How does it happen?

An infinite loop is a loop that repeats indefinitely because its stop condition never becomes False. This typically happens in a `while` loop if you forget to increment the counter variable inside the loop.

Q11. What are the three domains of AI?

The three domains of AI are: Data Sciences, Computer Vision (CV), and Natural Language Processing (NLP).

Q12. Name one gamified application of Computer Vision.

Google's "Quick Draw" game, which uses Computer Vision neural networks to recognize drawings made by users in real-time.

Q13. What is the purpose of the 4Ws problem canvas?

It is a framework used during the Problem Scoping stage of the AI Project Cycle to define a problem clearly by answering: Who is affected? What is the problem? Where does it occur? Why is it worth solving?

Q14. Define Data Literacy.

Data Literacy is the ability to read, understand, clean, analyze, visualize, and communicate with data to make informed decisions.

Q15. Differentiate between Data Privacy and Data Security.

Data Privacy is about user rights, consent, and control over who has access to personal data. Data Security is about protecting data from hackers, malware, and unauthorized access using technical tools like encryption and firewalls.

Q16. What is Tableau?

Tableau is an interactive data visualization tool that allows users to connect to data sources (like spreadsheets) and create dashboards, graphs, and charts.

Q17. What is the difference between Mean and Median?

The Mean is the calculated average of a set of numbers. The Median is the middle value in a sorted dataset. The median is less sensitive to extreme outliers than the mean.

Q18. What is a Generative Adversarial Network (GAN)?

A GAN is a machine learning architecture where two neural networks (a Generator that creates fake content and a Discriminator that flags fake content) compete against each other, helping the system learn to generate highly realistic data.

Q19. What is a deepfake?

A deepfake is synthetic media (images, audio, or video) generated by AI that replaces a person's likeness or voice with another, depicting them doing or saying things they did not.

Q20. What is an AI Hallucination?

An AI Hallucination is when a Generative AI model generates incorrect or false information confidently, presenting it as factual truth.

Projects, Virtual Field Visit & Portfolio

This part focuses on applying AI concepts to real-world social impact scenarios, particularly the UN Sustainable Development Goals (SDGs). Students complete project prototyping, virtual field visits, and maintain an activity portfolio.

IMPORTANT: CHOOSE ANY ONE

Students must complete **any one** of the following three options for Part D (15 marks):

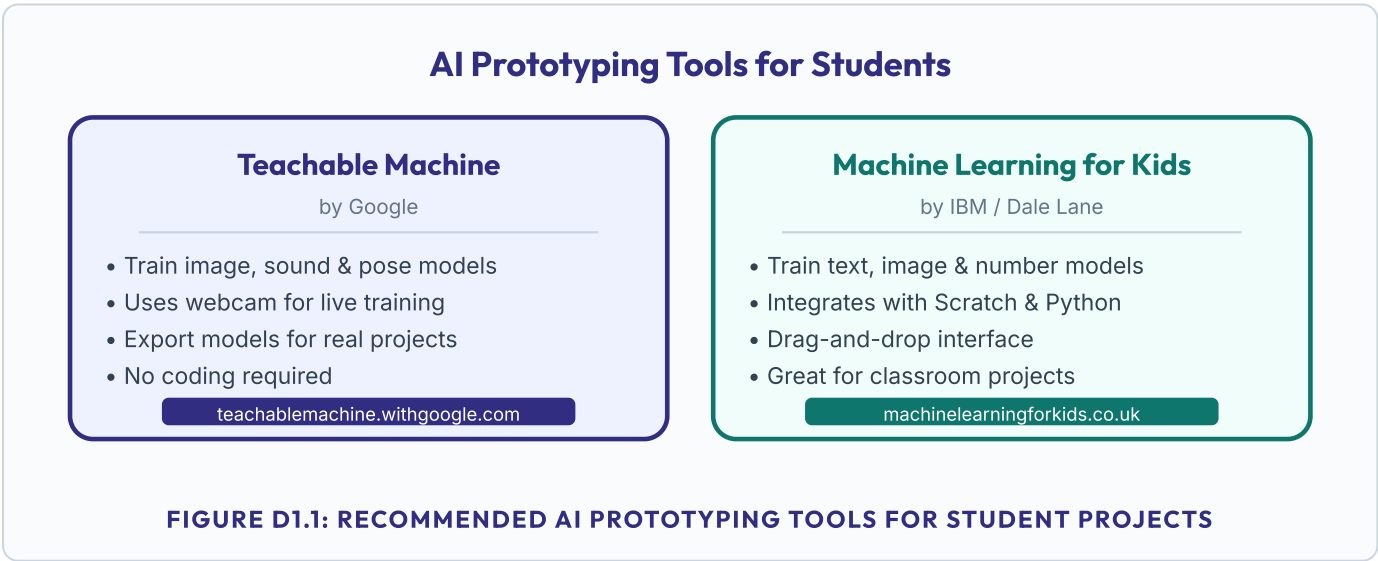
- Suggested Project** — Create an AI model OR complete an SDG-based project with data analysis.
- Suggested Field Visit** — Visit an AI-using company/industry (physical or virtual) and present a report.
- Student Portfolio** — Maintain a record of minimum 5 AI activities from the year.

D1.1 Social Impact Projects (AI for SDGs)

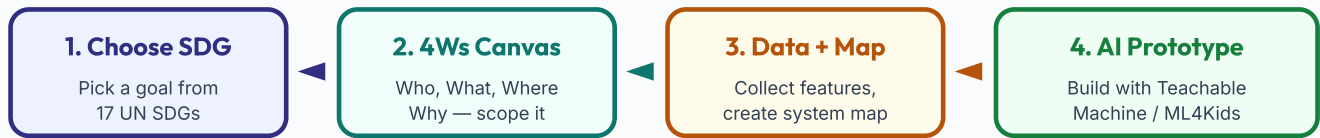
The UN has 17 Sustainable Development Goals (SDGs) to end poverty, protect the planet, and ensure prosperity. Students choose one SDG and follow these steps to propose an AI-enabled solution:

THE 4-STEP PROJECT FRAMEWORK

- Identify & Scope:** Choose a local issue aligned with an SDG (e.g., SDG 6: Clean Water, or SDG 15: Life on Land). Define the problem canvas using the 4Ws.
- Collect & Map:** List the data features needed. Create a **System Map** representing relationships (e.g., waste dumping rate vs. water cleanliness vs. local health cases).
- Analyze & Visualize:** Record data in a spreadsheet (**Google Sheets** or **Microsoft Excel** recommended) and plot bar charts or trendlines to demonstrate the severity of the problem.
- AI Prototype:** Propose an AI model to address the issue.
 - Example:* Training an image classifier on Google's **Teachable Machine** to sort waste automatically into biodegradable, recyclable, and hazardous bins.
 - Alternative Tool:* Use **Machine Learning for Kids** (<https://machinelearningforkids.co.uk/>) to build text, image, or number recognition models with a beginner-friendly drag-and-drop interface.



SDG Project Workflow — From Problem to Prototype



Present your findings to the class with data visualizations (bar charts, trendlines) and a working prototype or research proposal.

FIGURE D1.2: THE 4-STEP WORKFLOW FOR SDG SOCIAL IMPACT PROJECTS

D1.2 Virtual Field Visit

Students can conduct a physical or virtual field visit to a technology company or industry utilizing AI (e.g., smart warehouses, robotic assembly lines, or software companies using machine learning).

FIELD VISIT REPORT STRUCTURE

The presentation report must cover:

- **Name of Organization & Industry:** E.g., Robotic sorting center, automated dairy.
- **AI Technology Observed:** E.g., Computer Vision cameras inspecting fruit quality on conveyor belts.
- **Benefits Achieved:** Increased speed, reduced errors, and safer conditions for human workers.
- **Learnings:** Understanding how laboratory code is deployed in real-world environments.

D1.3 Student Activity Portfolio

Students maintain a record of at least 5 portfolio activities completed during the year:

1. **Letter to Future Self:** Reflecting on their career aspirations and how AI will impact jobs in the next 10 years.
2. **Smart Home Floor Plan:** A sketch showing sensors, smart hubs, and CV cameras, along with an explanation of the underlying logic.
3. **Future Job Advertisement:** Designing a job posting for a futuristic role (e.g., "AI Bias Auditor" or "Self-Driving Traffic Controller") listing required skills.
4. **SDG Action Plan:** The 4Ws canvas and system maps for their chosen social impact project.
5. **Research Work — AI for SDGs:** A research document exploring how AI is being used to address specific Sustainable Development Goals globally (e.g., AI in clean energy, AI for zero hunger).
6. **AI in Different Sectors:** A research paper analyzing AI applications in sectors like agriculture, healthcare, education, transportation, or gaming — with real-world examples.

PORTFOLIO TIP

You need a **minimum of 5 activities** in your portfolio. The list above has 6 options — pick any 5 (or do all 6 for extra credit!). Keep your portfolio organized in a folder or binder with clear headings and dates.

Equipment, System Specifications & Qualifications

This appendix lists the hardware and software specifications required to run the CBSE Class 9 AI curriculum, as well as the qualifications required for teachers and trainers.

A.1 Laboratory Equipment and System Configurations

To run the python sessions and data literacy visualization tools effectively, the computer laboratory should meet the following minimum specs (for a batch of 20 students, human-to-computer ratio of 2:1):

HARDWARE ITEM	MINIMUM REQUIREMENT / CONFIGURATION
Processor	Intel® Core™ i5-7300U Processor or equivalent (SYSmark® 2018 Rating of 750 or higher).
RAM	8GB DDR4 – 2400MHz or higher.
Storage	500 GB HDD – 7200 rpm or SSD equivalent.
Chassis / Form Factor	USFF (Ultra Small Form Factor) System chassis volume under one liter.
Graphics Card	Integrated graphics cards.
Webcam & Headset	Full HD Camera; Headphones with integrated microphone (for voice-recognition NLP sessions).
Display & I/O ports	18.5" LED Monitor with HDMI ports, built-in speaker, 4 USB 3.0 ports, 8-channel audio interface.
Wireless connectivity	Dual-band wireless connectivity (Minimum 800 Mbps), Bluetooth V4.2 or higher.
VPU (Vision Processing Unit)	Integrated VPU support to accelerate computer vision applications.

A.2 Software Configurations

- **Operating System:** Windows 10/11 or Ubuntu Linux.
- **Productivity Suite:** Google Workspace / G-Suite or MS Office.
- **Internet Browser:** Google Chrome (required for online AI games and Tableau Public).
- **Python Environment:** Anaconda Distribution (includes Jupyter Notebook and Spyder IDE) or Python IDLE version 3.8+.
- **Intel OpenVINO Toolkit:** Installed for AI neural network optimization and computer vision lab modules.

A.3 Instructor Qualifications & Standards

Teachers and trainers responsible for delivering the CBSE AI curriculum must meet the following criteria:

Qualifications

- Diploma in Computer Science / Information Technology OR
- Bachelor's Degree in Computer Applications (BCA) / B.Sc. Computer Science / B.Sc. IT OR
- Graduation in any stream with a Post Graduate Diploma in Computer Applications (PGDCA) OR
- DOEACC 'A' Level Certificate.

Experience and Skills

- Minimum of 1 year of work experience in a similar teaching or technical job role.
- Ability to communicate effectively in English and the local language.

- Knowledge of classroom safety, equipment handling, health, and hygiene standards.
- Age limit: 18–37 years (at the time of appointment; standard relaxation rules apply).

STANDARD SELECTION & TRAINING PROCEDURE

- **Written Test:** Assesses technical knowledge in AI, Python programming, and data literacy.
- **Interview:** Panel interview evaluating interest, aptitude, and communication skills.
- **Mock Class Demo:** Demonstrating teaching style and classroom management.
- **Induction Training:** Selected instructors undergo a mandatory 20-day training program on vocational pedagogy and the NSQF framework.
- **In-Service Training:** Instructors complete 5 days of training annually to stay updated on new AI technologies and curriculum changes.